



SDN and NFV Overview



Foreword

- The open ecosystem of the computing industry brings booming development of multiple fields, such as Commercial Off-the-Shelf (COTS), operating system, virtualization, middleware, cloud computing, and software applications. The network industry is also seeking transformation and development. Software Defined Networking (SDN) and Network Functions Virtualization (NFV) are mainly used.
- This course aims to help engineers understand the development of SDN and NFV and introduce Huawei SDN and NFV solutions.



Objectives

- Upon completion of this course, you will be able to:
 - Describe the development of SDN and NFV.
 - Understand basic principles of OpenFlow.
 - Understand Huawei SDN solution.
 - Understand the standard NFV architecture.
 - Understand Huawei NFV solution.



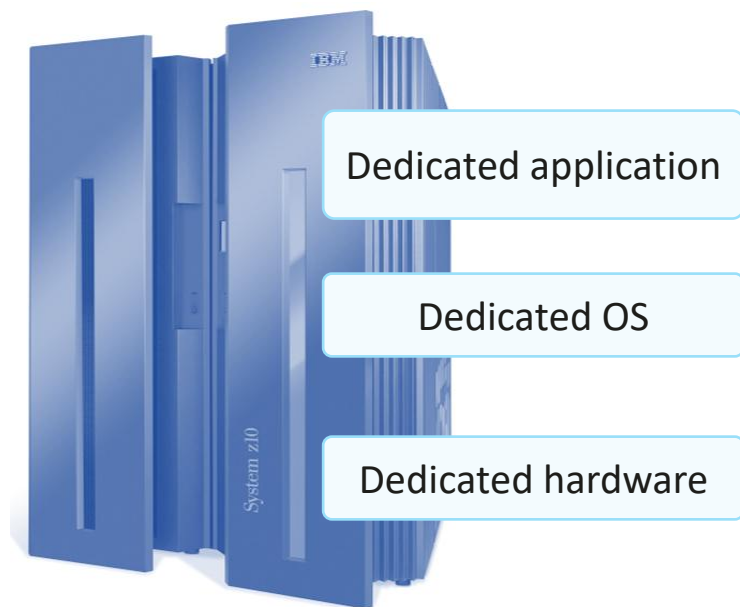
Contents

- 1. SDN Overview**
2. NFV Overview



Evolution of the Computer Era

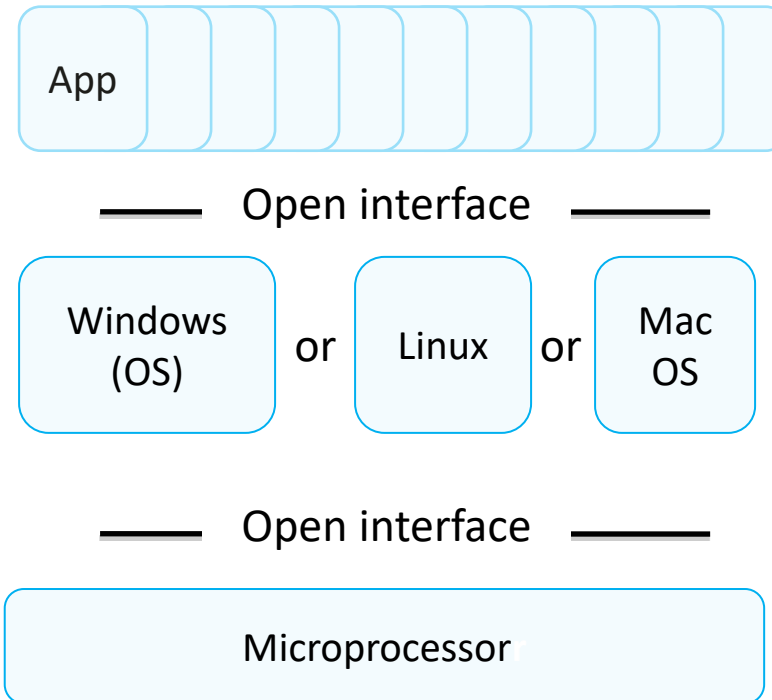
Mainframe



Vertical integration and closed interfaces
Small-scale industry applications



PC (compatible)



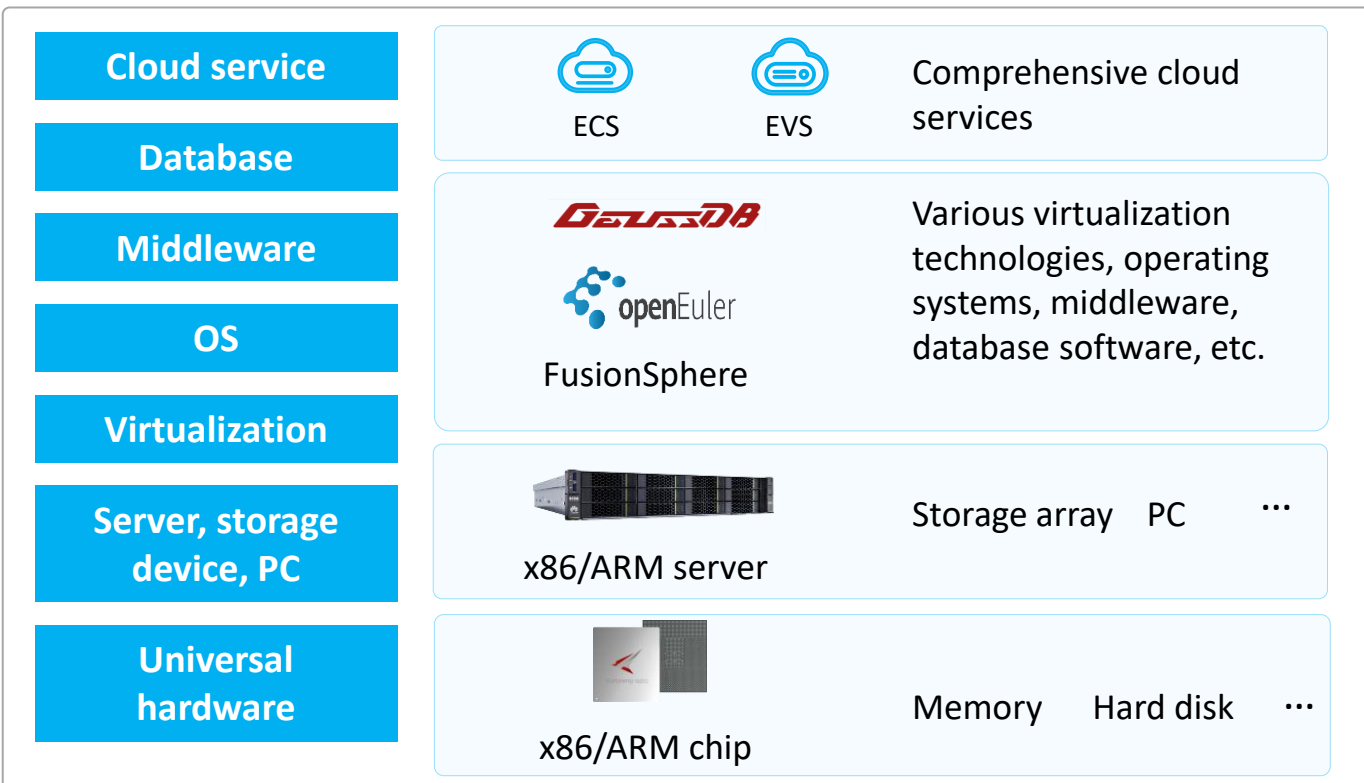
Horizontal integration and open interfaces
Large-scale application across industries



Network Industry Development: Implications from the IT Industry

- The transformation of the IT industry has triggered the thinking of the network industry. The industry has proposed the SDN concept and has made attempts to put SDN into commercial use, aiming to make networks more open, flexible, and simple.

Computing Industry Openness Promotes Ecosystem Development



What About Network Industry Changes



Network application

SDN controller

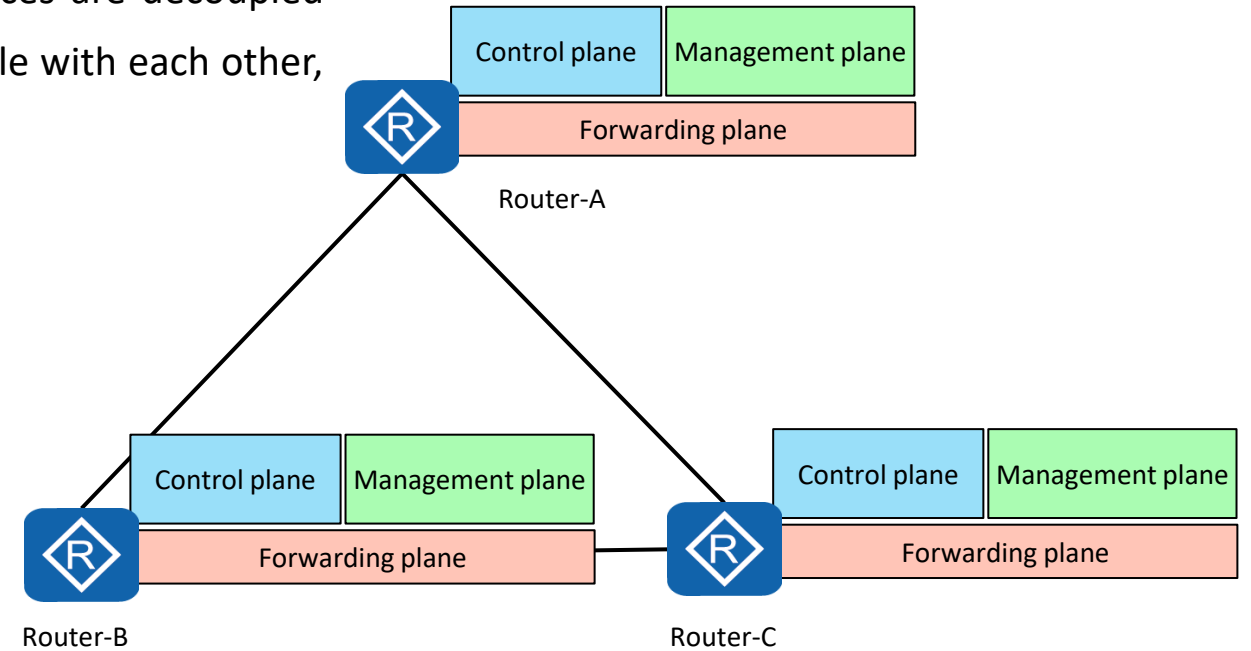
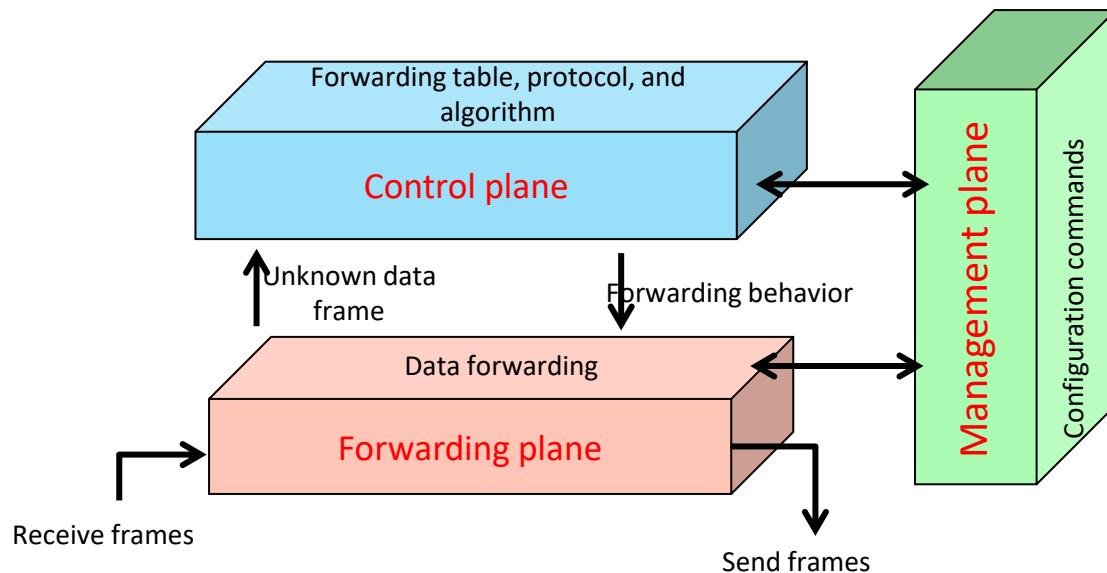
Hardware network device

- Does the network industry build a hierarchical and open ecosystem according to the computing industry?



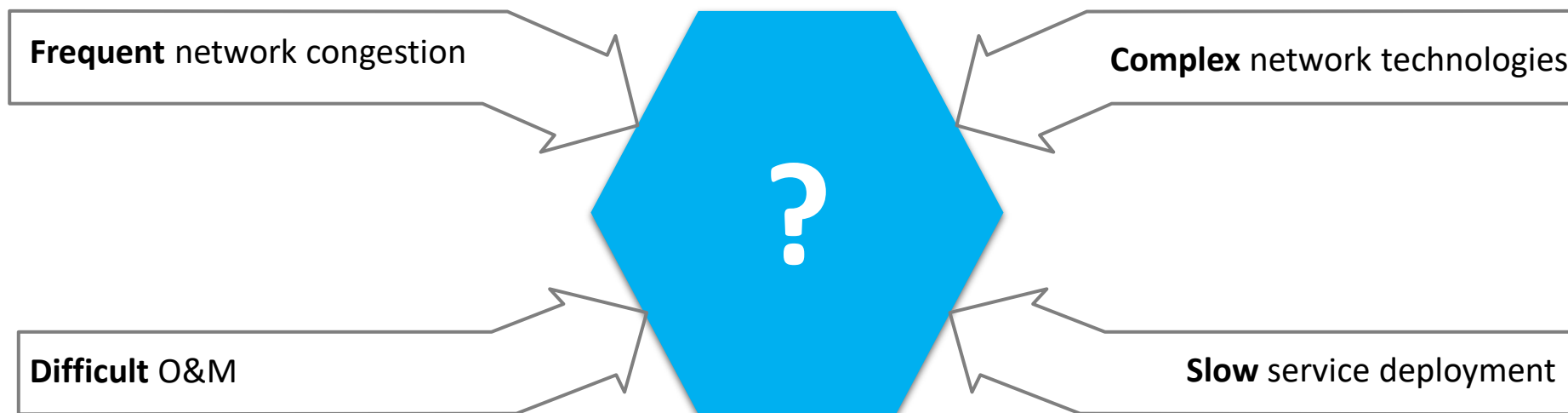
Current Situation of the Network Industry: Typical IP Network - Distributed Network

- The typical IP network is a distributed network with peer-to-peer control. Each network device has independent forwarding, control, and management planes. The control plane of a network device exchanges packets of a routing protocol to generate an independent data plane to guide packet forwarding.
- The advantage of a typical IP network is that network devices are decoupled from protocols, devices from different vendors are compatible with each other, and network convergence is ensured in fault scenarios.





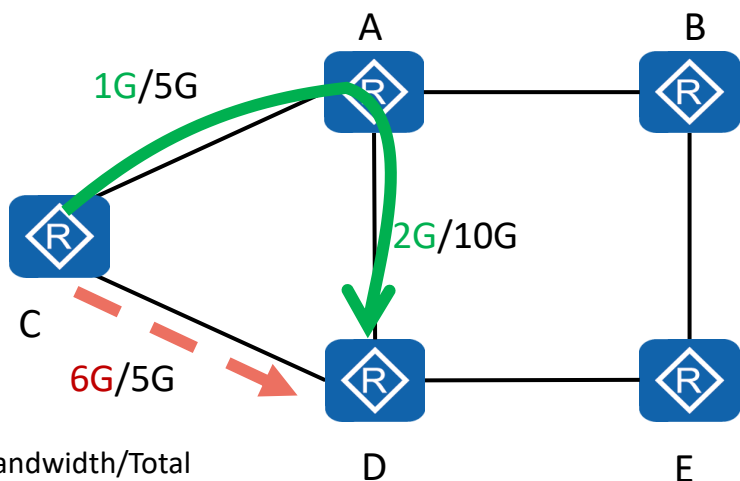
Thinking in the Network Field: Problems Faced by Typical Networks





Frequent Network Congestion

Problem and Solution of Bandwidth-based Route Selection

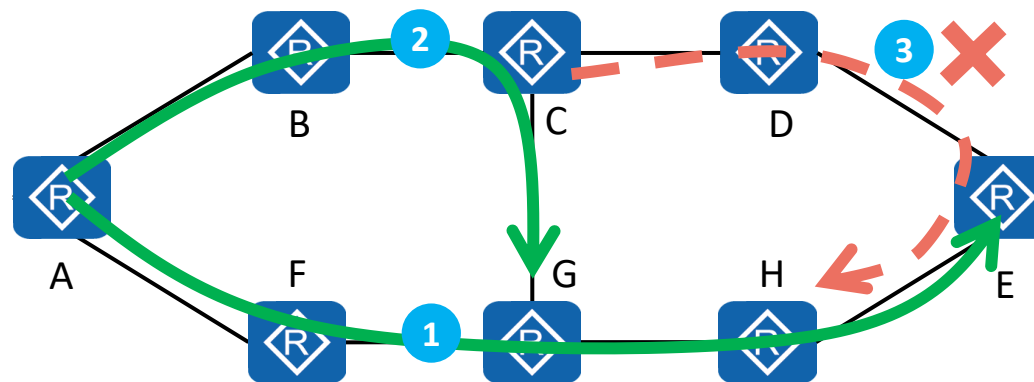


Used bandwidth/Total
bandwidth

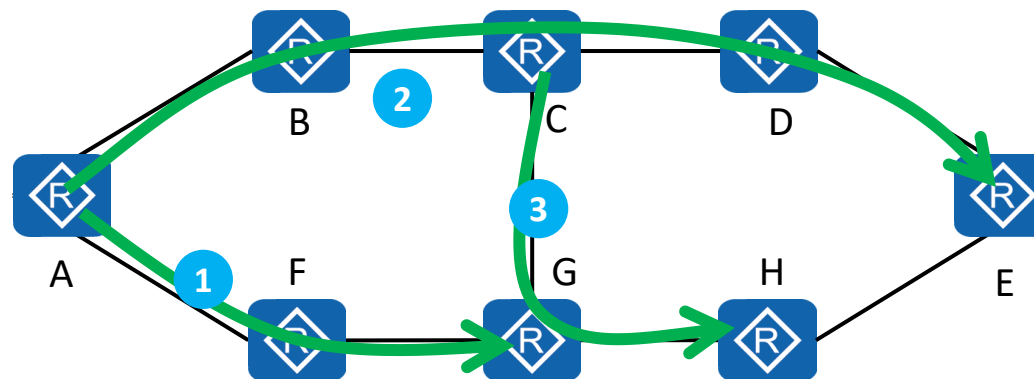
The network computes forwarding paths based on bandwidth. The link from router C to router D is the shortest forwarding path. The volume of service traffic from router C to router D exceeds the bandwidth, causing packet loss. Although other links are idle, the algorithm still selects the shortest path for forwarding. The optimal traffic forwarding path is C-A-D.

Problem and Solution of Tunnel Establishment Based on Fixed Sequence

Tunnels are established in sequence: 1. A-E; 2. A-G; 3. C-H. Tunnel 3 fails to be established due to insufficient bandwidth.



Global path calculation and optimal tunnel path adjustment:



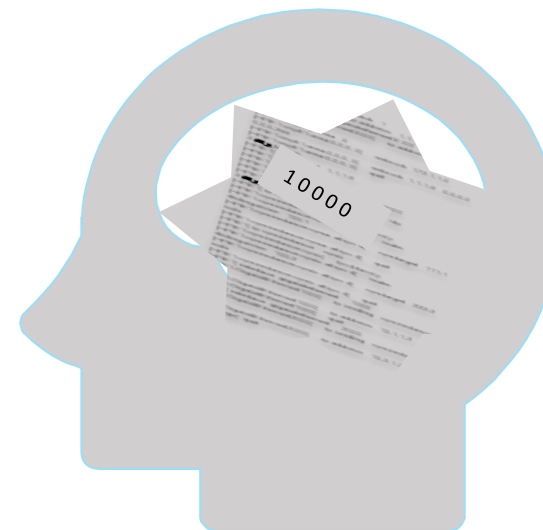
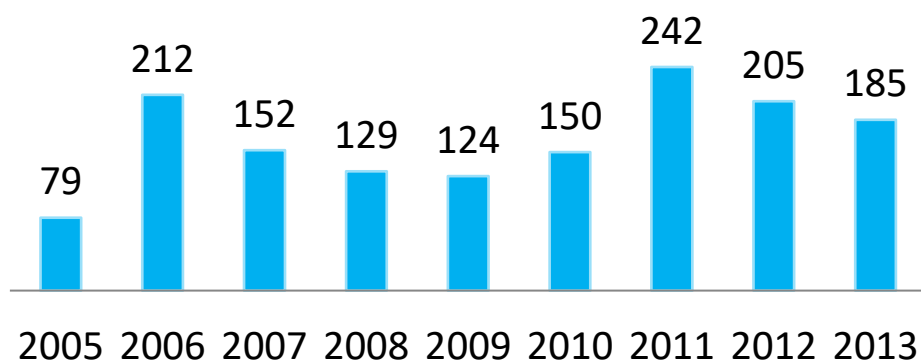


Complex Network Technologies

Many network protocols: Network technology experts need to learn many RFCs related to network devices. Understanding the RFCs takes a long time, and the number of RFCs is still increasing.

Difficult network configuration: To be familiar with devices of a specific vendor, you need to master tens of thousands of commands. Additionally, the number of commands is still increasing.

RFC increase trends





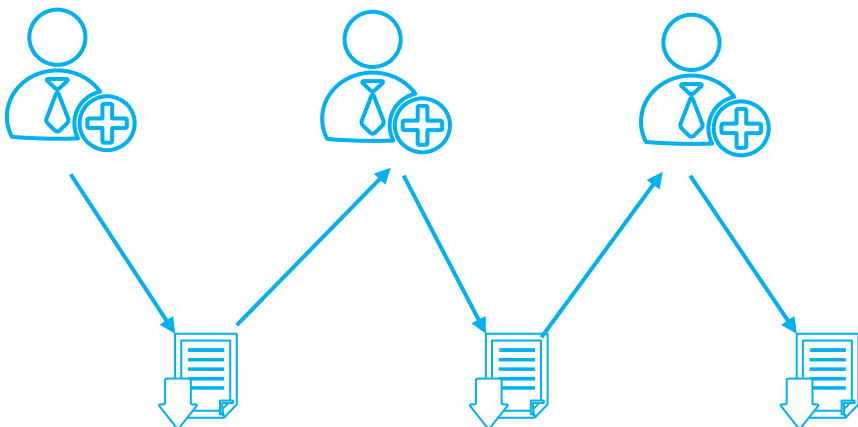
Difficulty in Locating and Analyzing Network Faults

Difficult to Spot Faults

Manual fault
identification

Manual packet obtaining
for locating faults

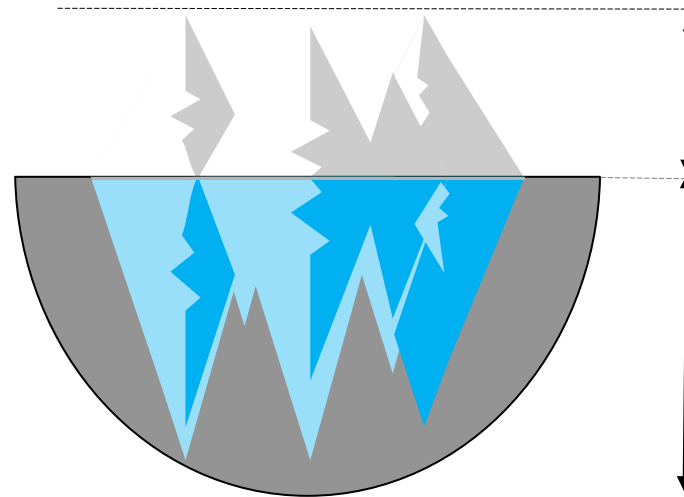
Manual fault diagnosis



- Traditional O&M networks rely on manual fault identification, location, and diagnosis.
- More than 85% of network faults are found only after service complaints. Problems cannot be proactively identified or analyzed.

Difficult to Locate Faults

Abnormal flows account for 3.65% of all flows on the network.



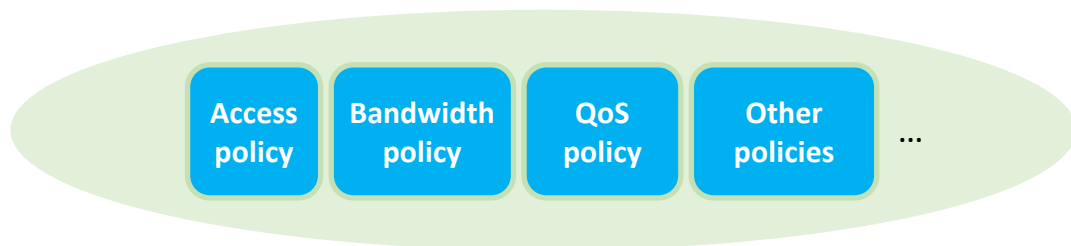
The network faults that are found upon user complaints are just the tip of the iceberg.

- Traditional O&M only monitors device indicators. Some indicators are normal, but user experience is poor. There is no correlated analysis of users and networks.
- According to data center network (DCN) statistics, it takes 76 minutes to locate a fault on average.



Slow Network Service Deployment

Network policy



Complex and inflexible network policy changes:

Network policies cannot be defined by user. Policy changes are complex and cannot be flexibly adjusted.

IP address-based, fixed location, and CLI-based configuration

Service network

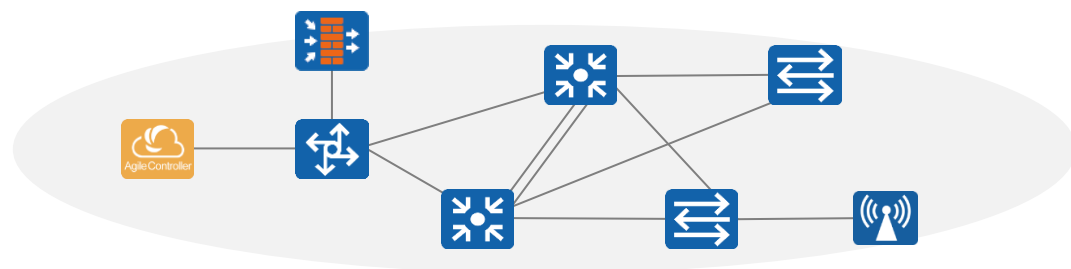


Long service deployment period:

New service deployment involves E2E device configuration modification.

End-to-end configuration using commands

Physical network



Low physical network deployment efficiency:

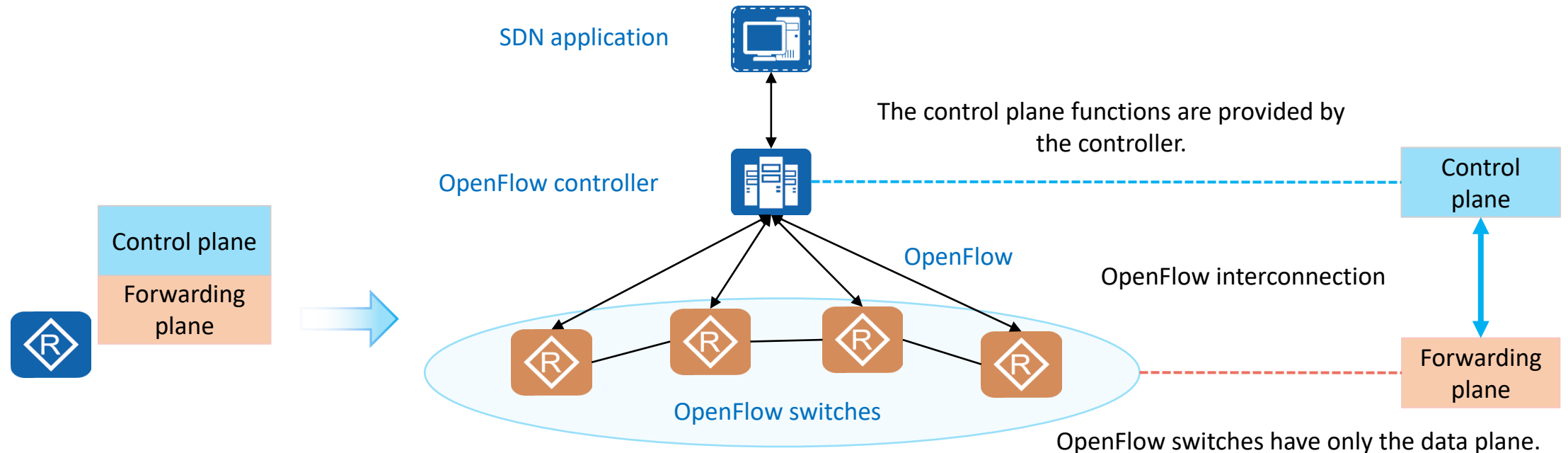
The physical network does not support zero touch provisioning (ZTP).

Command line-based configuration by device



SDN Origin

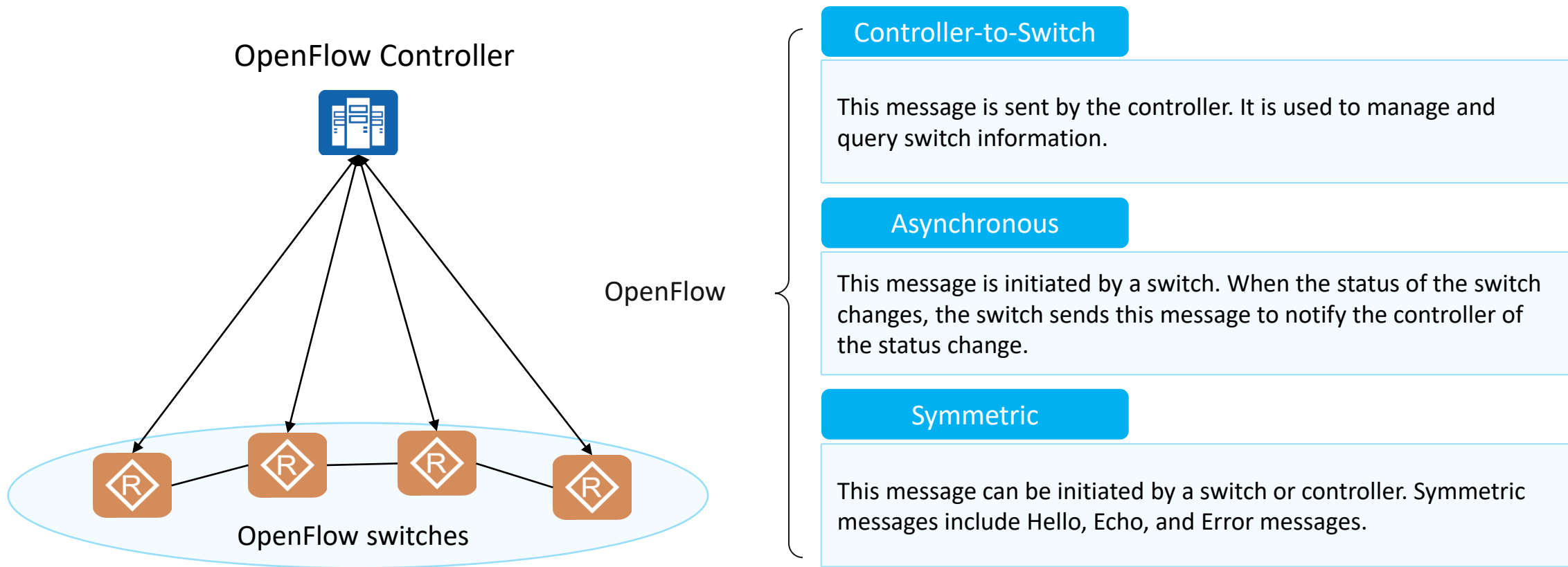
- SDN was developed by the Clean Slate Program at Stanford University as an innovative new network architecture. The core of SDN is to separate the control plane from the data plane of network devices to implement centralized control of the network control plane and provide good support for network application innovation.
- SDN has three characteristics in initial phase: forwarding-control separation, centralized control, and open programmable interfaces.





Basic Concepts of OpenFlow

- OpenFlow is an SBI protocol between a controller and a switch. It defines three types of messages: Controller-to-Switch, Asynchronous, and Symmetric. Each message contains more subtypes.





Flow Table Overview

- OpenFlow switches forward packets based on flow tables.
- Each flow entry includes the Match Fields, Priority, Counters, Instructions, Timeouts, Cookie, and Flags. The Match Fields and Instructions are key fields for packet forwarding.
 - The Match Fields is a field against which a packet is matched and can be customized.
 - The Instructions field indicates OpenFlow processing when a packet matches a flow entry.

Match Fields	Priority	Counters	Instructions	Timeouts	Cookie	Flags
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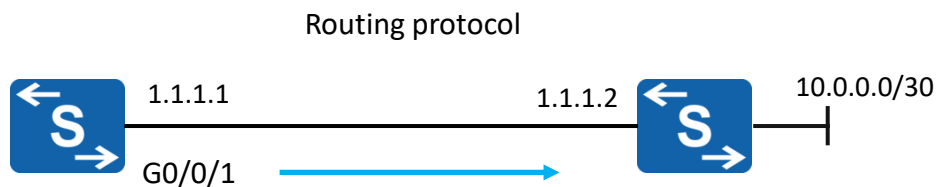
Flow table fields can be customized. The following table is an example.

Ingress Port	Ether Source	Ether Dst	Ether Type	VLAN ID	VLAN Priority	IP Src	IP Dst	TCP Src Port	TCP Dst Port
3	MAC1	MAC2	0x8100	10	7	IP1	IP2	5321	8080



Comparison Between Forwarding Modes

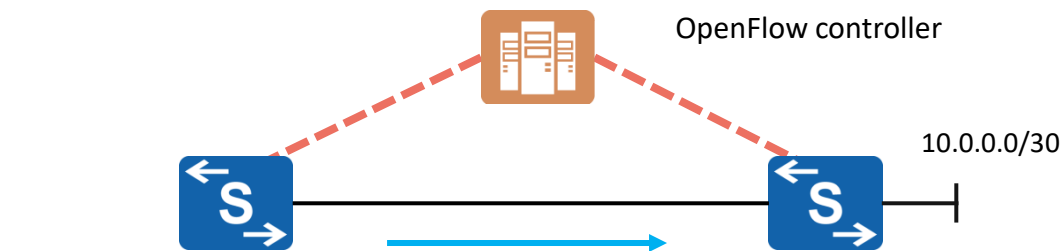
Typical Routing Protocol: Packet Forwarding Based on Routing Tables



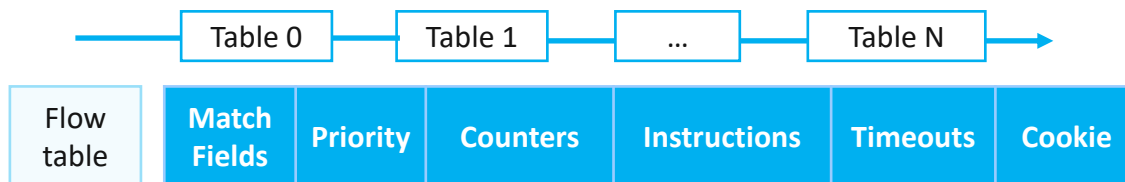
Routing table	Destination Network	Protocol	Next Hop	Outbound Interface
	10.0.0.0/30	OSPF	1.1.1.2	G0/0/1

- In typical cases, network devices query routing tables to guide traffic forwarding.
- Entries in a routing table are calculated by running a routing protocol between network devices.
- The length of the routing table is fixed. Network devices forward packets based on the longest match rule. A network device has only one routing table.

OpenFlow: Packet Forwarding Based on Flow Tables



Flow table matching process:



- OpenFlow is a network protocol. Switches running OpenFlow forward traffic based on flow tables.
- Flow tables are calculated by the OpenFlow controller and then delivered to switches.
- A flow table has variable length and defines various matching and forwarding rules. A network device has multiple flow tables.



Essential Requirements of SDN

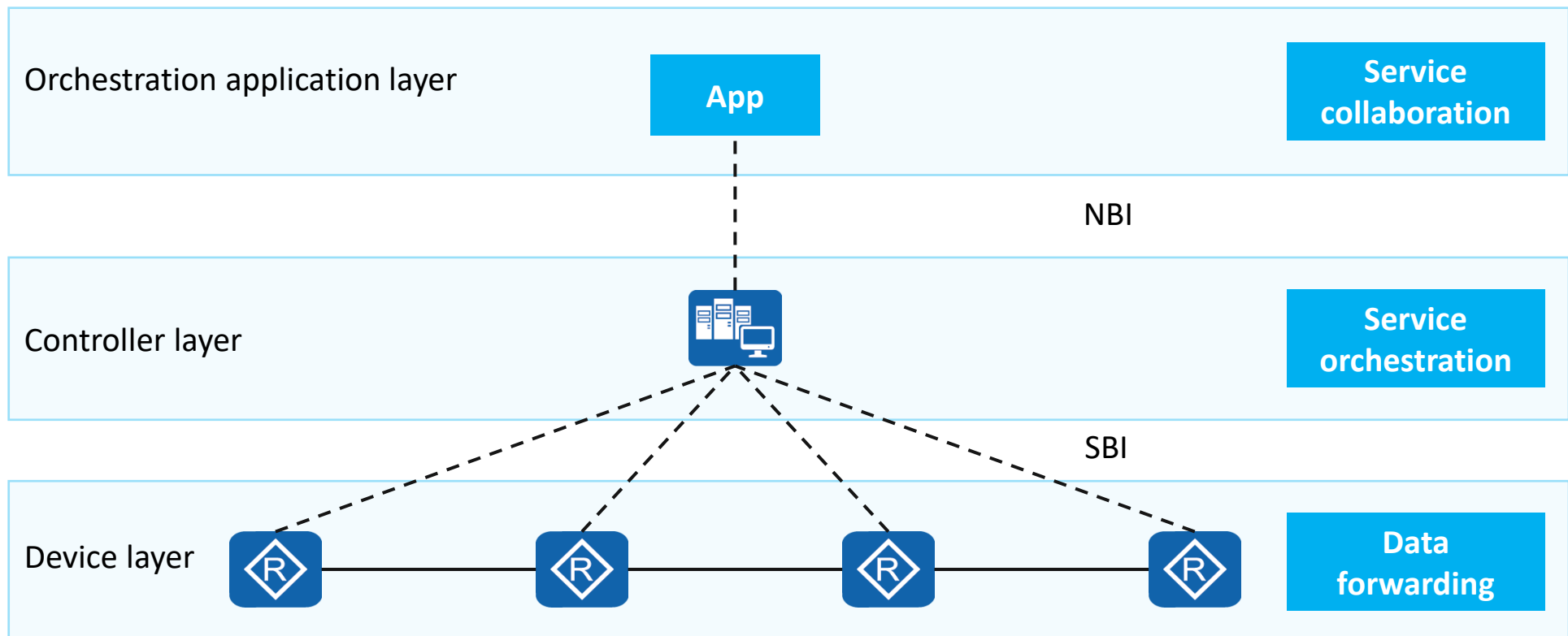
- The essence of SDN is to make networks more open, flexible, and simple. It builds a centralized brain for a network and implements fast service deployment, traffic optimization, or network service openness through centralized control in the global view.
- SDN has the following benefits:
 - Provides centralized management, simplifying network management and O&M.
 - Shields technical differences, simplifies network configuration, and reduces O&M costs.
 - Offers automatic optimization, improving network utilization.
 - Deploys services rapidly, shortening the service rollout time.
 - Builds an open network, supporting open and programmable third-party applications.

SDN transforms network architecture.



SDN Network Architecture

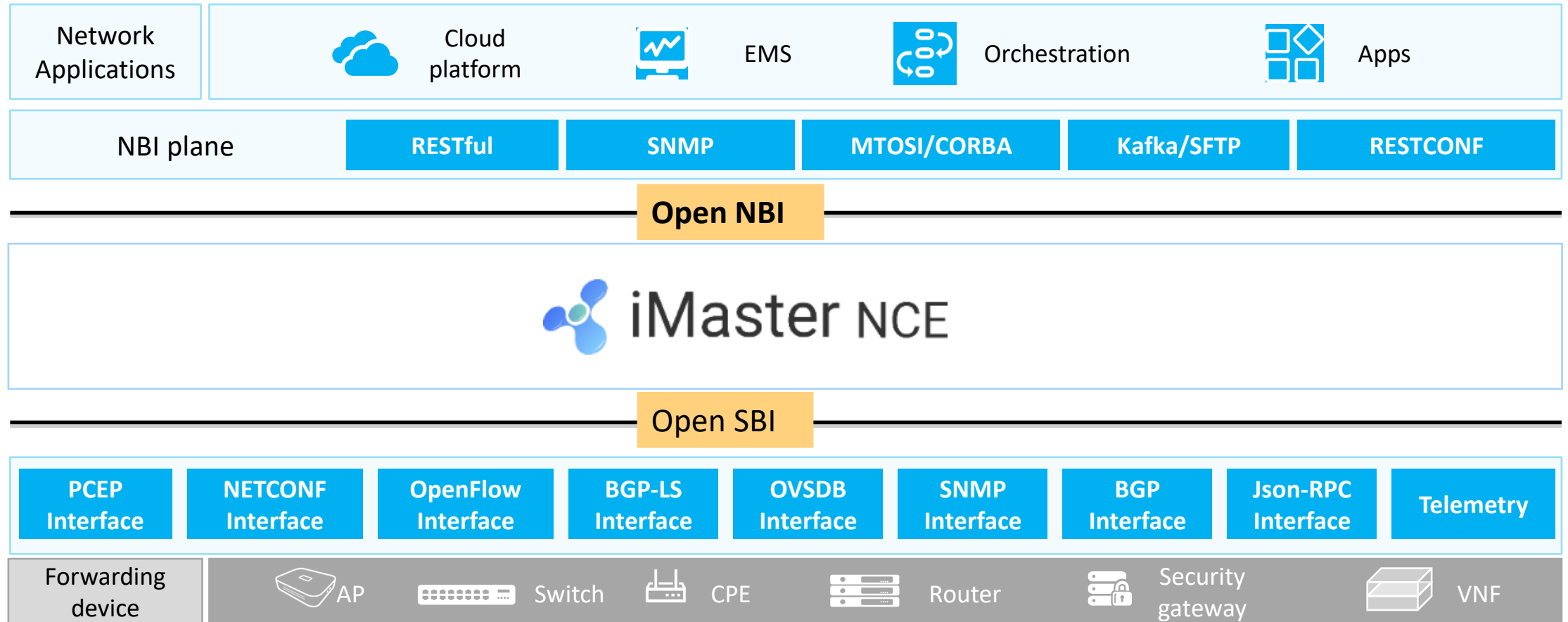
- The SDN network architecture consists of the orchestration application layer, controller layer, and device layer. Different layers are connected through open interfaces. From the perspective of the controller layer, SBIs oriented to the device layer and NBIs oriented to the orchestration application layer are distinguished. OpenFlow is one of SBI protocols.





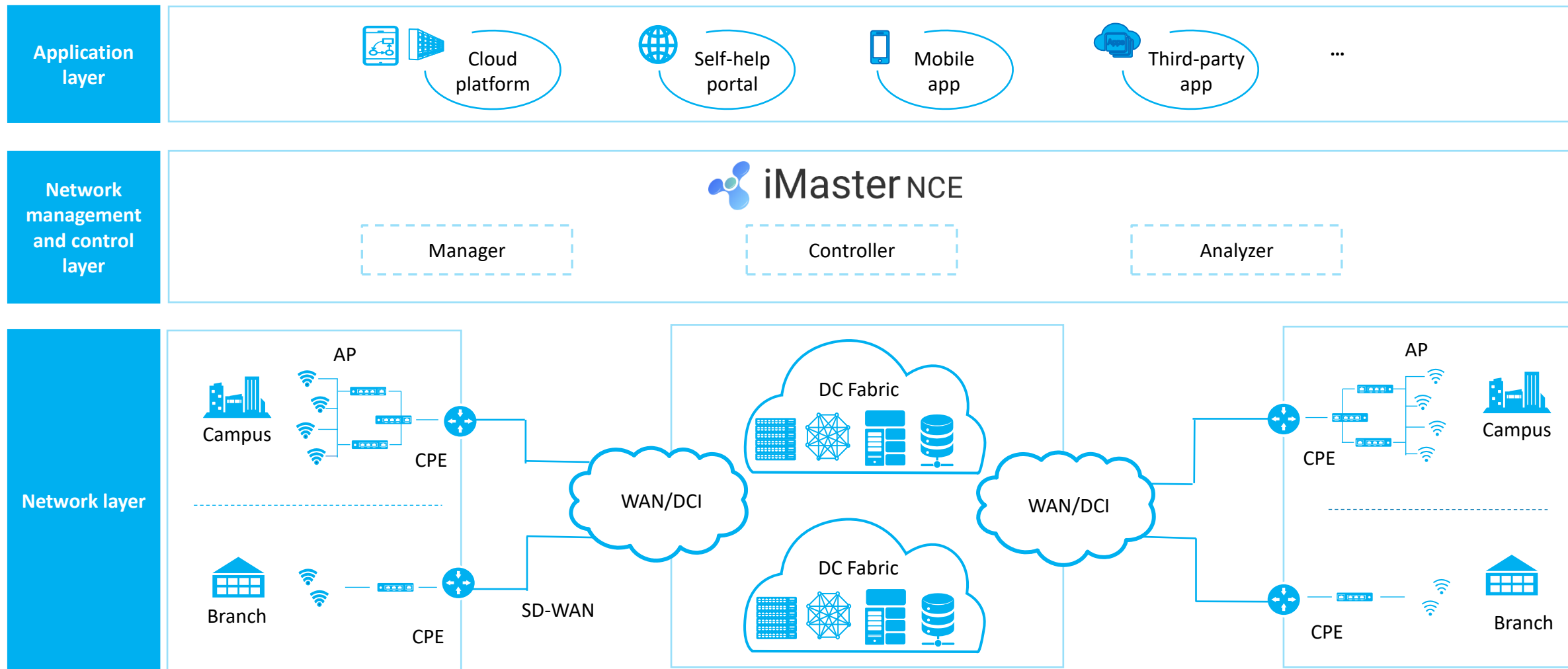
Huawei SDN Network Architecture

- Huawei SDN network architecture supports various SBIs and NBIs, including OpenFlow, OVSDB, NETCONF, PCEP, RESTful, SNMP, BGP, JSON-RPC, and RESTCONF interfaces.





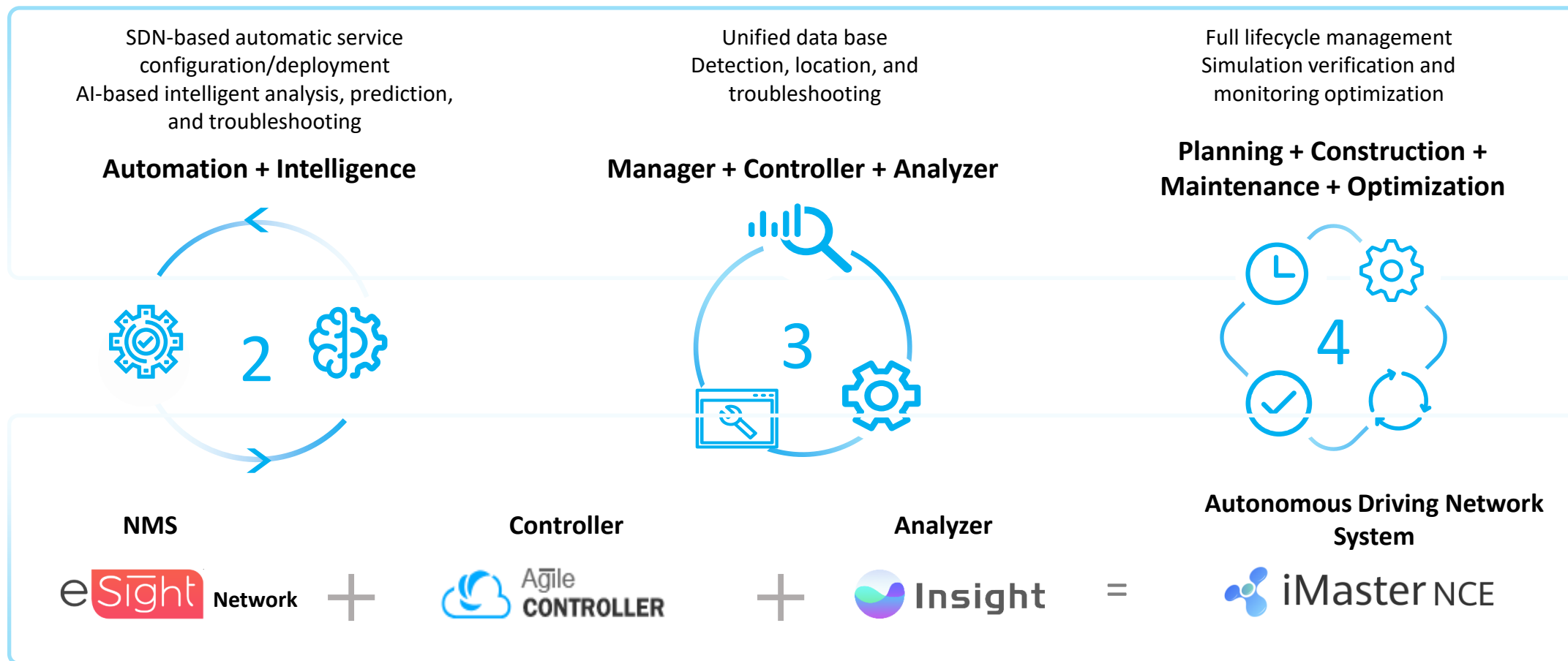
Huawei SDN Solution - Integrating Management, Control, and Analysis to Build an Intent-Driven Network





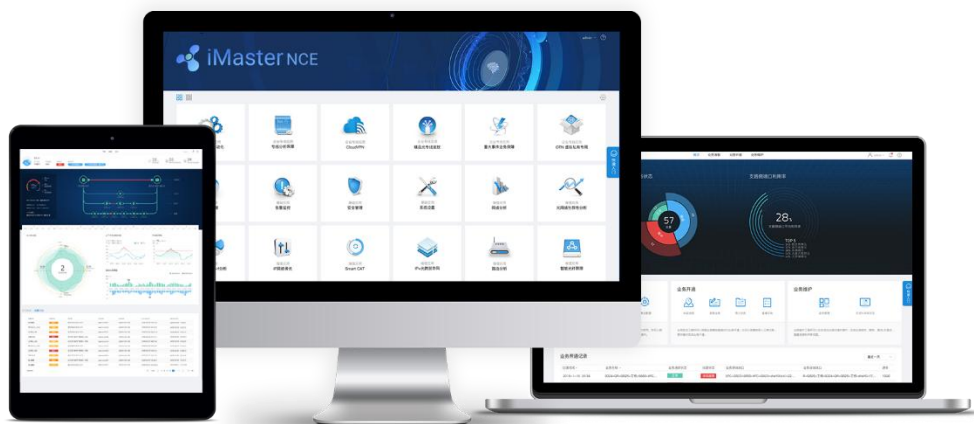
Introduction to iMaster NCE

- Huawei iMaster NCE is the industry intelligent network automation platform that integrates management, control, analysis, and AI capabilities.





iMaster NCE Application



DC

iMaster NCE-Fabric *



Enterprise
campus

iMaster NCE-Campus *



SD-WAN

iMaster NCE-WAN



IP WAN

iMaster NCE-IP



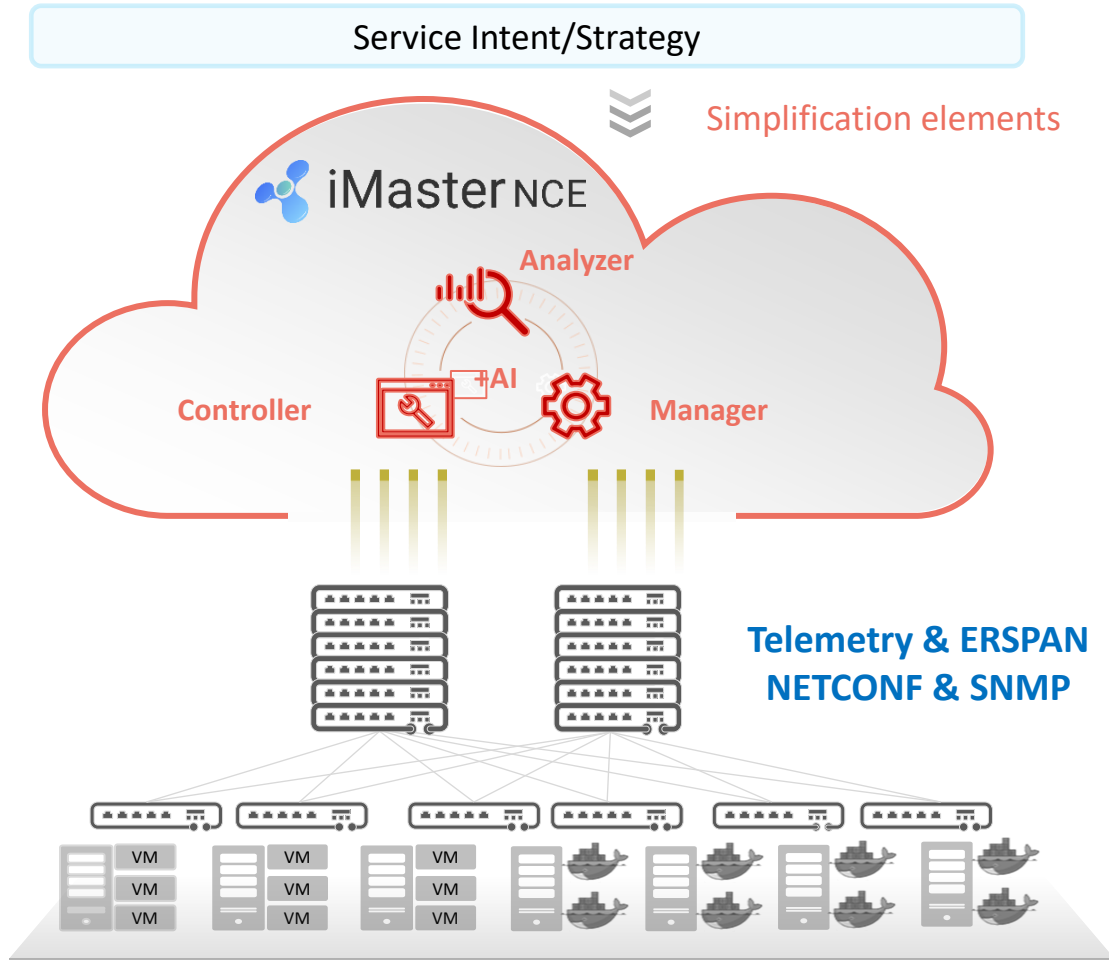
WAN
transmission

iMaster NCE-T

* iMaster NCE-Fabric and iMaster NCE-Campus are introduced in this document.

Huawei CloudFabric DCN Autonomous Driving Network Solution

- Based on iMaster NCE-Fabric, DCNs provide full-lifecycle services from planning, construction, O&M, to optimization.



Integrated planning and construction:

- The planning tool interconnects with iMaster NCE-Fabric to implement integrated planning and construction.
- Zero Touch Provisioning (ZTP)

Simplified deployment

- Service intent self-understanding and conversion
- Network change simulation and evaluation, eliminating human errors

Intelligent O&M:

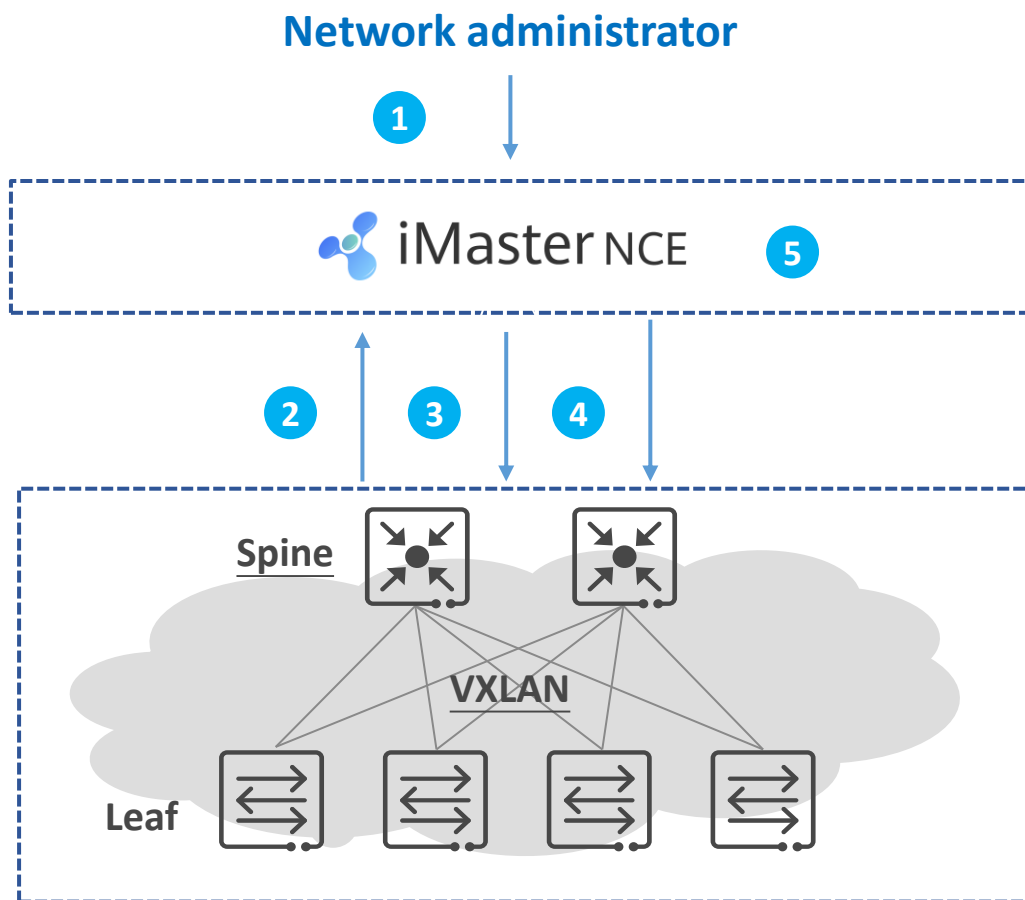
- Rapid fault detection and location based on knowledge graph and expert experience
- Fast fault rectification based on expert experiences and simulation analysis

Real-time optimization:

- AI-Fabric-oriented local traffic inference and online model training and optimization
- User behavior prediction and resource optimization suggestions



Simplified ZTP Deployment



ZTP deployment process:

1. The network administrator clicks the icon on iMaster NCE to start the ZTP task.
2. A device automatically obtains an IP address to access iMaster NCE.
3. iMaster NCE determines the device role (spine or leaf node), delivers configurations such as the management IP address, SNMP configuration, and NETCONF configuration to online devices, and manages the devices through the management IP address.
4. iMaster NCE globally delivers interconnection configurations as well as OSPF or BGP configurations.
5. The device goes online successfully, and the administrator views network-wide information on iMaster NCE.

Note: The DC uses the spine-leaf architecture.



Network Intent Self-understanding and Fast Service Deployment

Network design

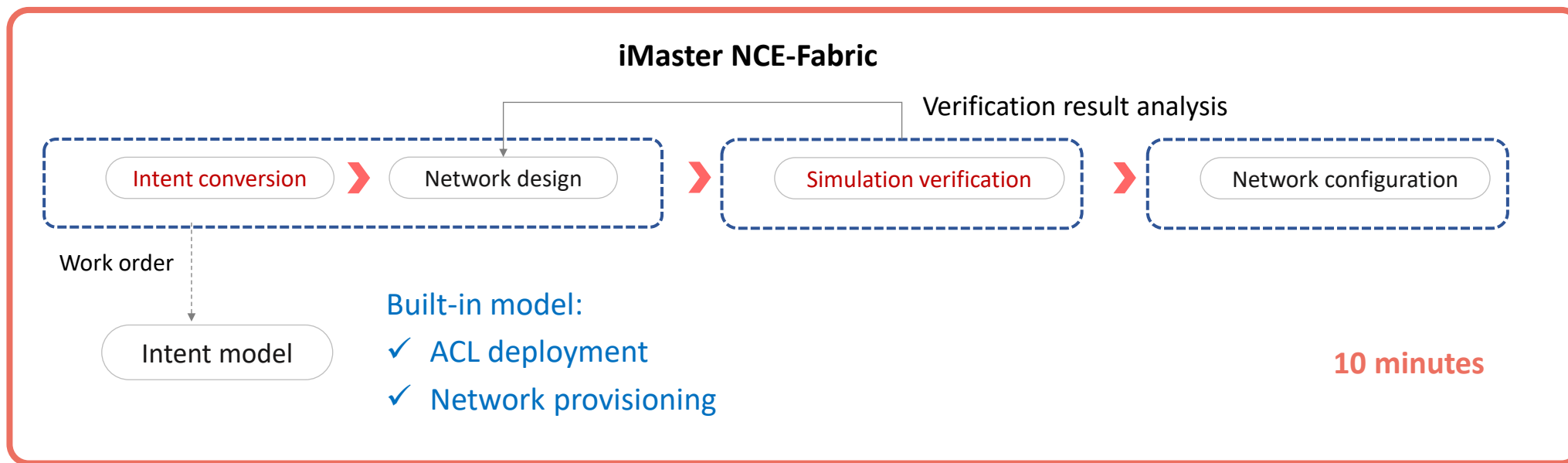
2 to 3 days

Configuration delivery

10 minutes

Service verification

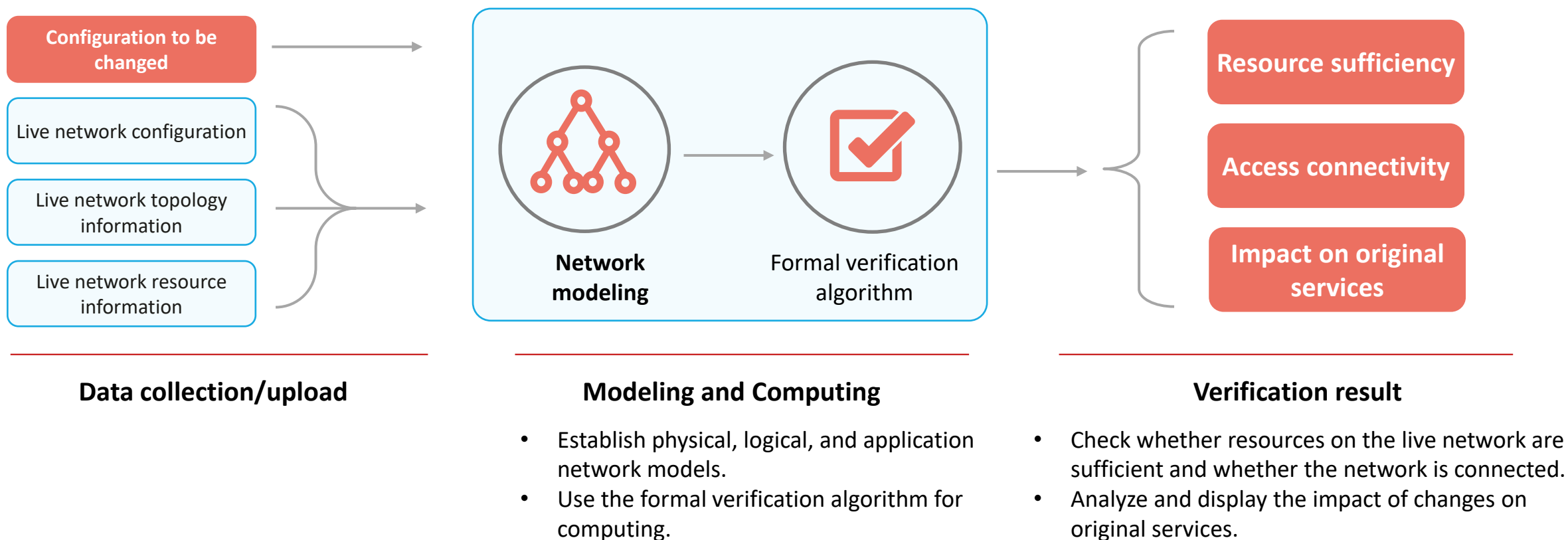
1 to 2 days



Huawei iMaster NCE-Fabric supports automatic and fast deployment of virtualization, cloud computing, and container networks.

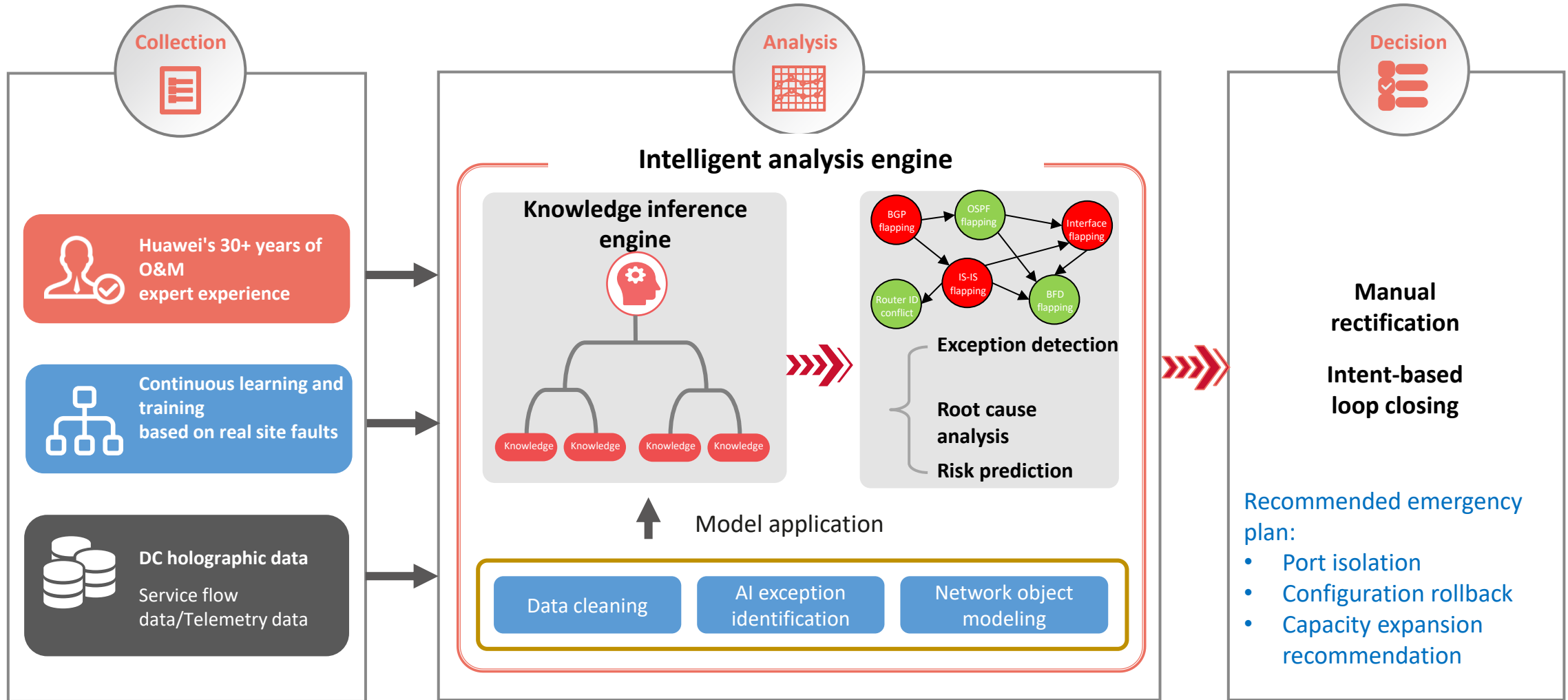


Network Change Simulation and Change Risk Prediction



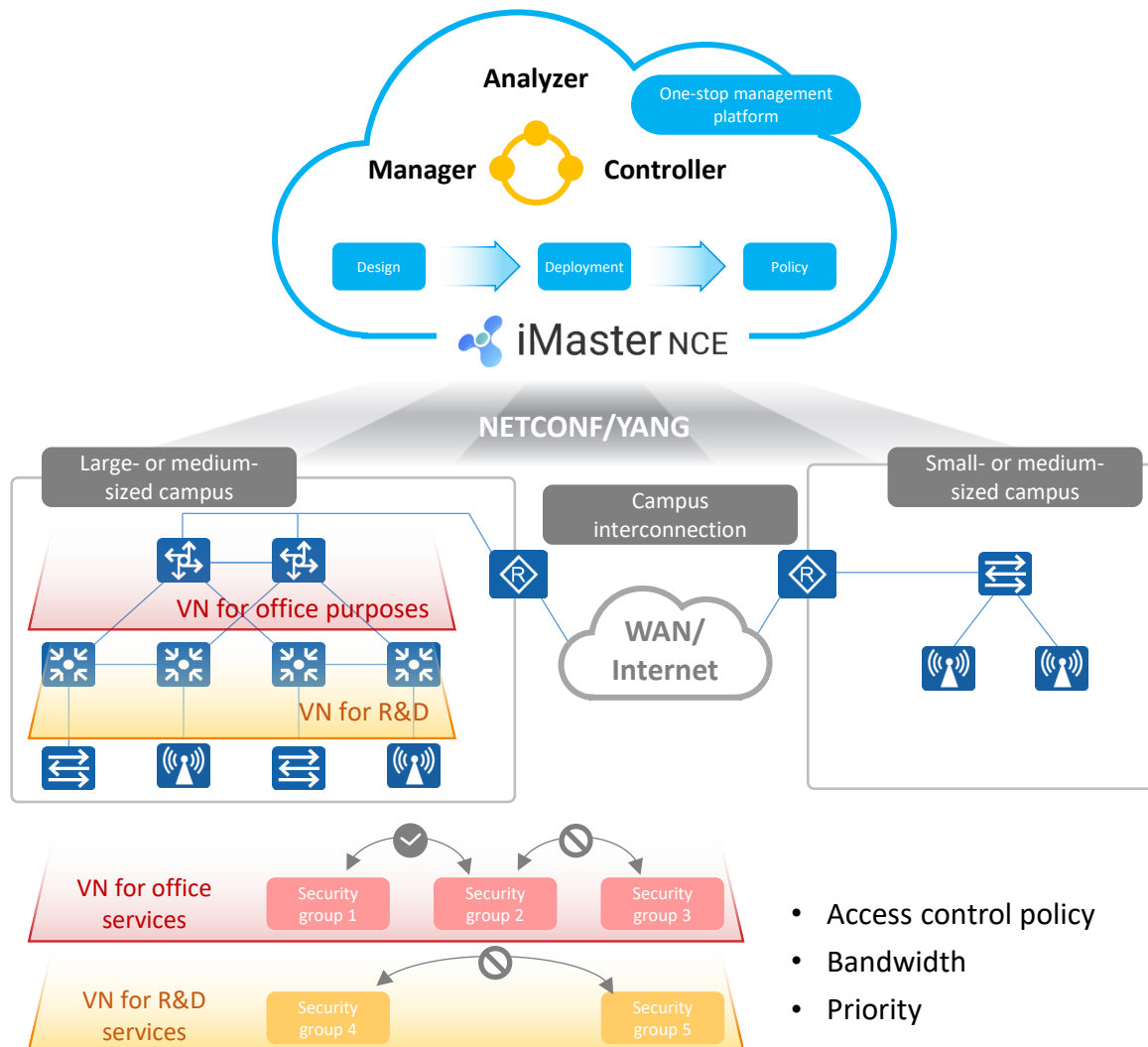


AI-powered Intelligent O&M for DCNs





Huawei CloudCampus Autonomous Driving Network Solution



Fast network deployment, improving deployment efficiency by 600%

- Device plug-and-play: simplified device deployment, scenario navigation, and template-based configuration
- Simplified network deployment: Network resource pooling, multi-purpose network, and automatic service provisioning

Fast service provisioning, improving user experience by 100%

- Free mobility: GUI-based policy configuration, allowing users to access the network anytime and anywhere without changing the roaming permission and user experience
- Intelligent terminal identification: Anti-spoofing for terminal access, with an intelligent terminal identification accuracy of over 95%
- Intelligent HQoS: Application-based scheduling and shaping, and refined bandwidth management, ensuring service experience of key users

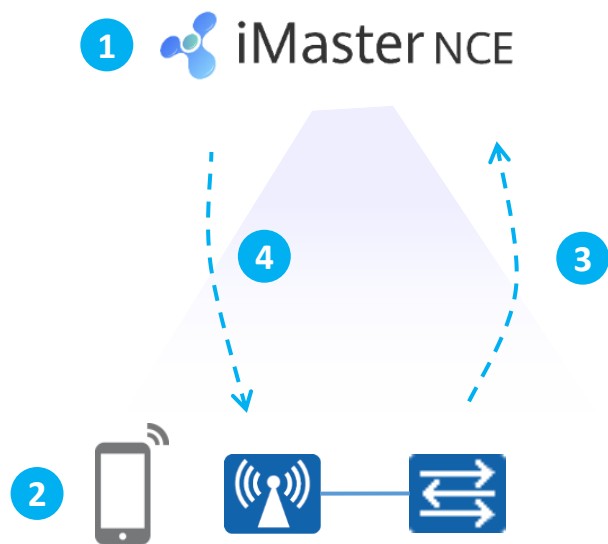
Fast intelligent O&M, improving network performance by over 50%

- Real-time experience visualization: Telemetry-based network experience visualization at each moment, for each user, and in each area
- Precise fault analysis: Proactively identifying 85% of typical network issues and providing suggestions, and comparing and analyzing real-time data to predict faults
- Intelligent network optimization: Predictive optimization of wireless networks based on historical data, improving network-wide performance by over 50% (Source: Tolly Certification)



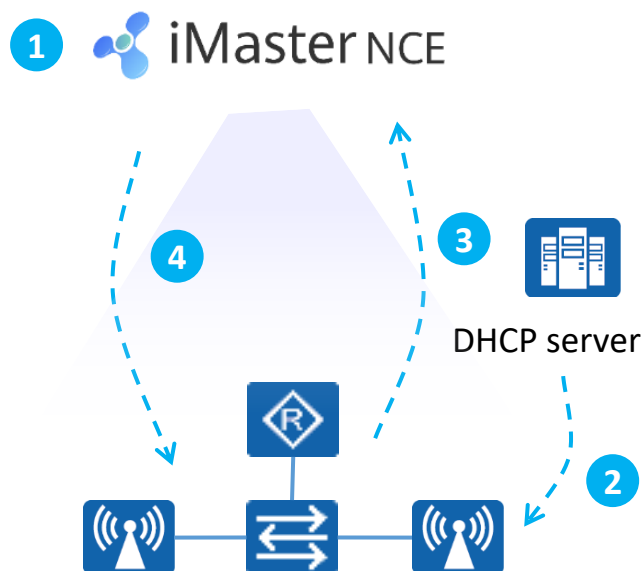
Device Plug-and-Play

Deployment by Scanning Bar Codes



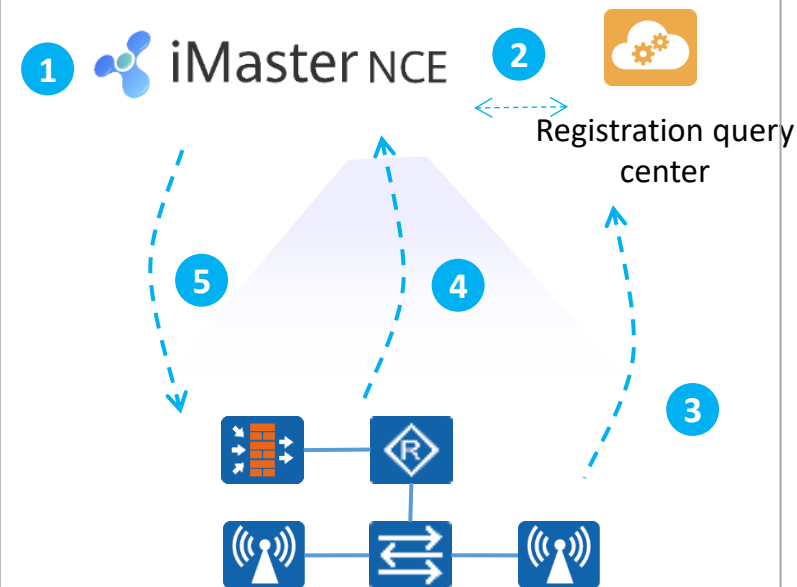
1. Pre-configuration
2. Deployment by scanning bar codes
3. Automatic device registration and login
4. Automatic configuration delivery

DHCP-based Deployment



1. Pre-configuration
2. Obtaining registration information through the DHCP server
3. Automatic device registration and login
4. Automatic configuration delivery

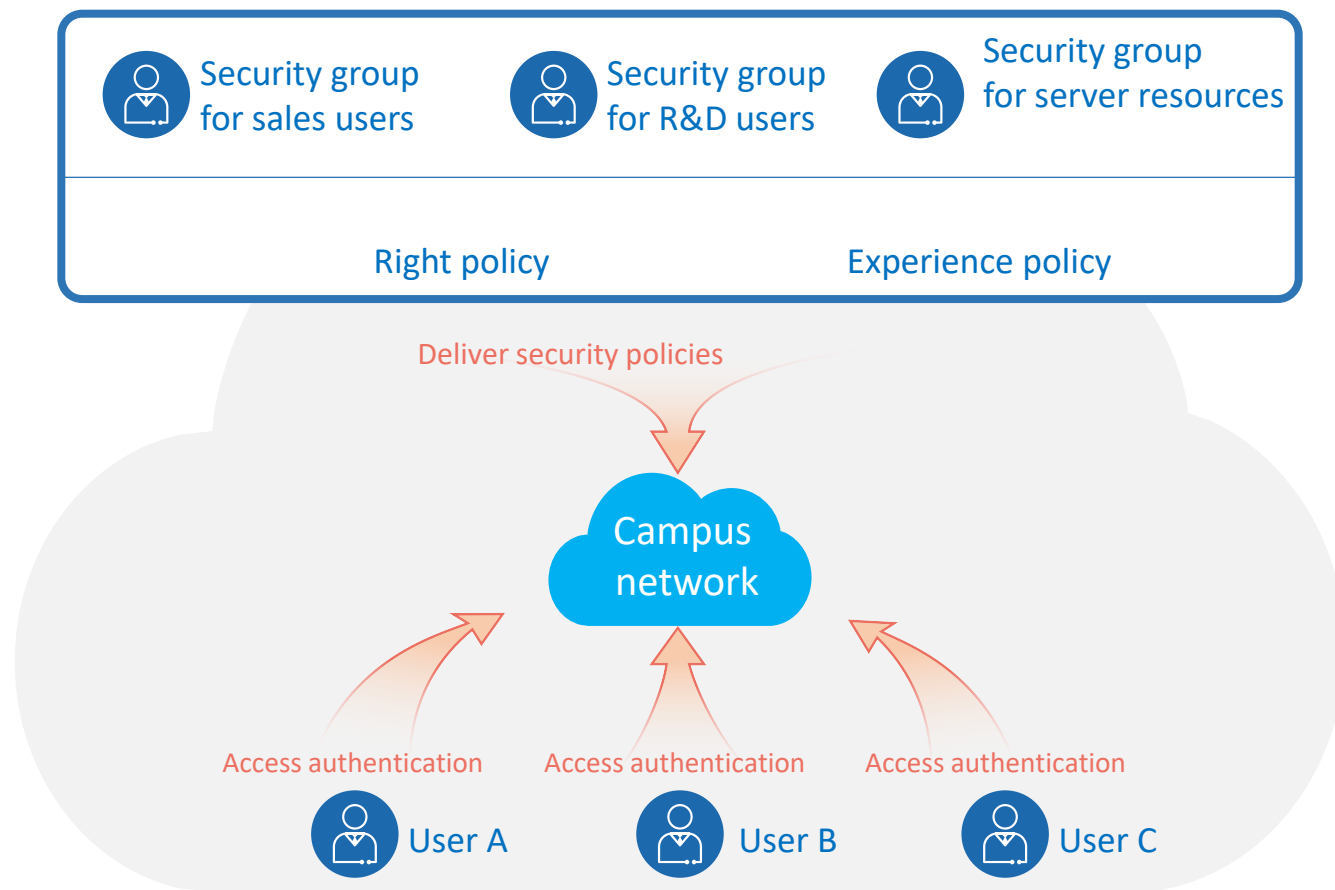
Deployment Through the Registration Query Center



1. Pre-configuration
2. Information synchronization
3. Obtaining registration information through the registration query center
4. Automatic device registration and login
5. Automatic configuration delivery

Free Mobility: Policy Management Based on Security Groups

- Free mobility: Enables users to have consistent network rights and security policies regardless of their locations and IP addresses.

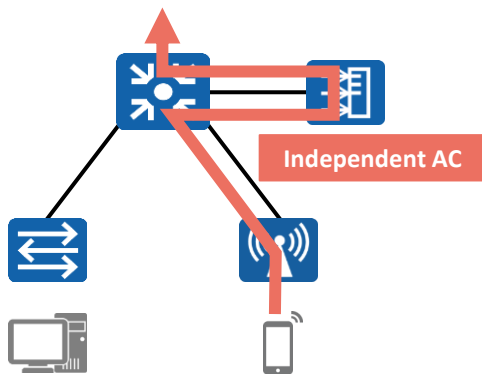


- 1 Use security groups. A security group is a group of users for which the same security policy is used.
- 2 Define security group-based permission control policies and user experience policies and deliver the policies to network devices.
- 3 A security group is authorized to a user after the user passes access authentication.
- 4 After user traffic enters a network, network devices enforce policies based on the source and destination security groups of the traffic.



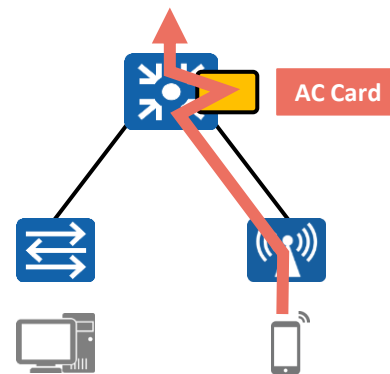
Wired and Wireless Convergence

WLAN Construction Mode 1: Standalone AC



- This mode poses a wireless traffic bottleneck and increases faulty nodes.
- Wired and wireless management is independent.
- Wired and wireless authentication points are separated.

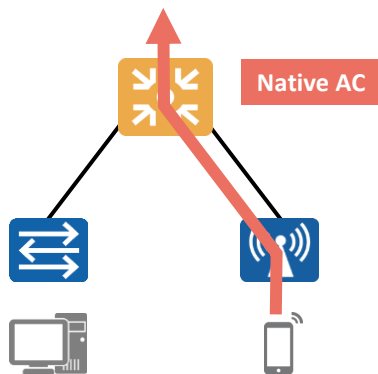
WLAN Construction Mode 2: AC Card



- An AC card is installed on a switch to provide AC functions.
- Hardware-level convergence.

Wired and wireless authentication point separation, distributed policy control, separation of control and data traffic forwarding, and troubleshooting and management difficulties.

Wired and Wireless Convergence (Native AC)



The switch integrates the AC function, eliminating wireless traffic forwarding bottlenecks and reducing fault nodes. Wired and wireless devices are centrally managed.

- Unified management and converged forwarding of wired and wireless services
- Converged management for wired and wireless users and gateway convergence
- Converged authentication points for wired and wireless access
- Unified wired and wireless policy execution



Intelligent Terminal Identification, Ensuring Secure Access

DC

Enterprise
campus

Requirements and Challenges

A university

50+ types of smart terminals

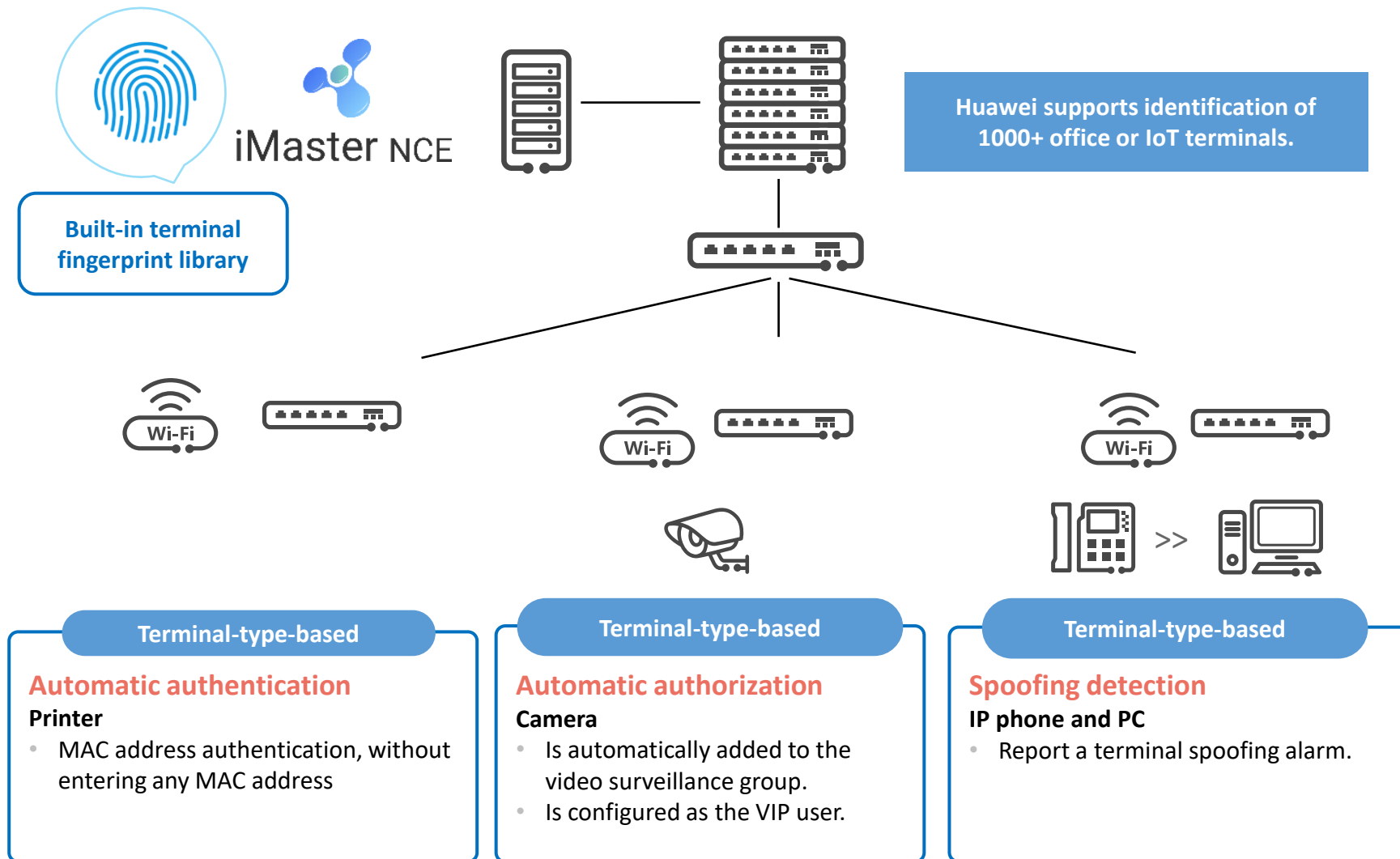
Terminal information is
collected by IT departments
of colleges:
MAC address collection is
difficult and error-prone.

An enterprise

100+ authentication faults
reported per day

It is difficult to locate access
spoofing.

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HQoS: User- and Application-based QoS Policy

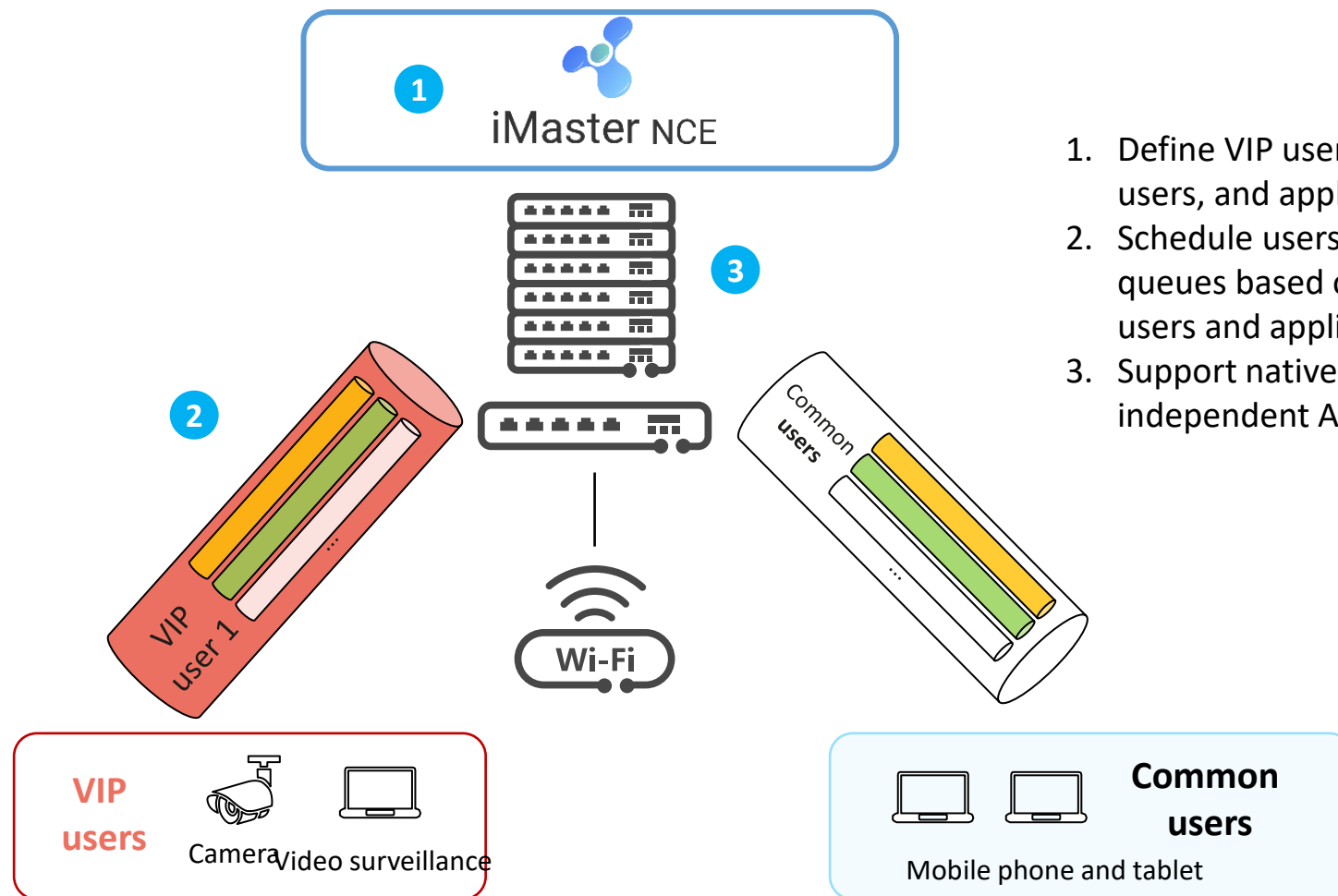
User- and application-based QoS policies ensure experience of high-priority users and applications

Requirements and Challenges

Traditional QoS policies are invalid for video services.

(Example) Building surveillance scenario:
Wireless video services of common users increase, occupying a large number of network resources and causing network congestion.

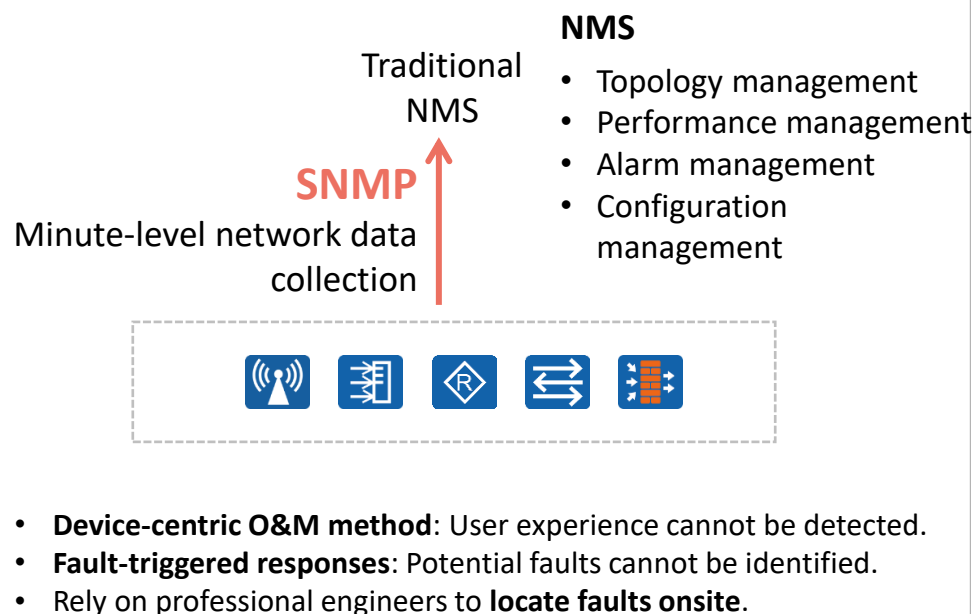
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AI-Powered Intelligent O&M of Campus Networks

As-Is: Device-Centered Network Management



To-Be: User Experience-Centered AI-Powered Intelligent O&M



Telemetry
Second-level network data collection

Intelligent network analyzer

- Visualized user experience management
- User journey playback
- Potential fault identification
- Root cause identification
- Predictive network optimization



Experience visualization: Telemetry-based second-level data collection, visualized experience of each user and each application in real time

Minute-level potential fault identification and root cause location

- Identify potential faults based on dynamic baselines and big data correlation analysis.
- Accurately locate root causes of faults through KPI association analysis and protocol tracing.

Predictive network optimization: AI is used to intelligently analyze the AP load trend and implement predictive optimization on the wireless network.

The efficiency is improved by using algorithms. With scenario-based continuous learning and expert experience, intelligent O&M frees O&M personnel from complex alarms and noises, making O&M more automated and intelligent.



AI-Powered Intelligent Radio Calibration

Traditional radio calibration cannot achieve expected results

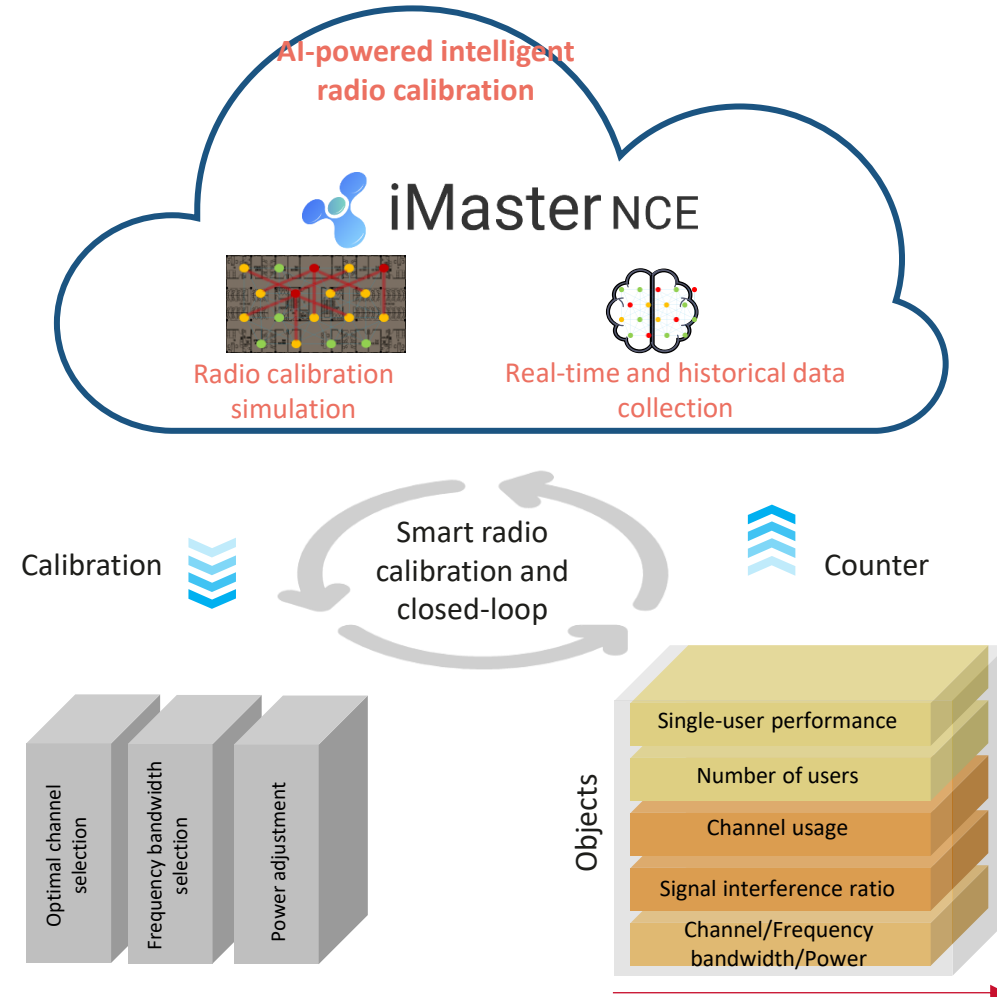
Phase 1: Manual Calibration

Based on engineers' experience, the calibration is time-consuming and error-prone, and the calibration result is unstable.



Phase 2: Automatic Calibration

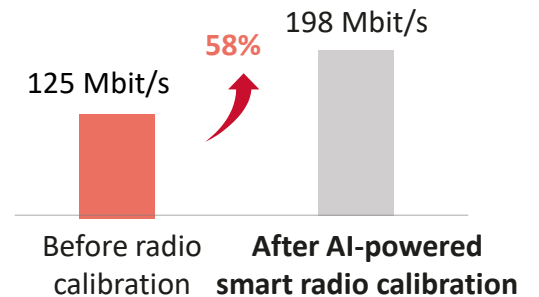
Calibration based on real-time radio interference does not consider device load changes, and the calibration result cannot be ensured.



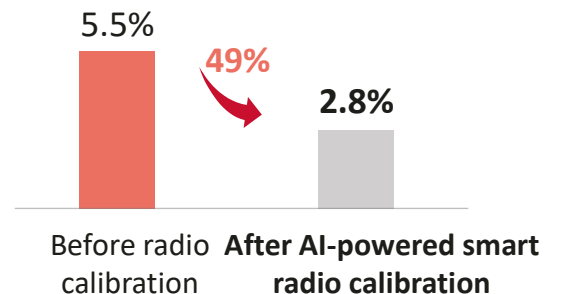
Tested and verified by authoritative organizations



Average downlink rate per terminal:



Average Wi-Fi channel interference





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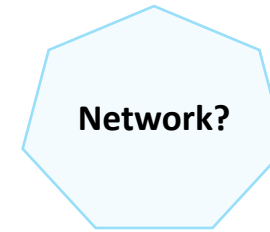
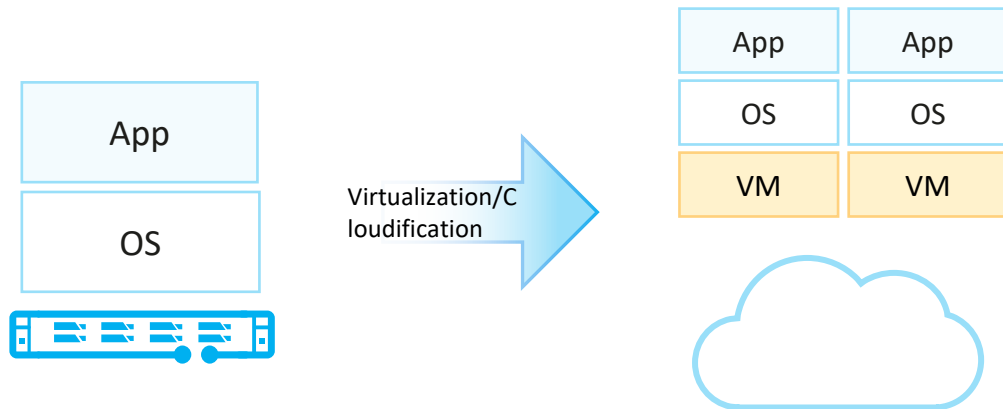


NFV Background: Thinking from IT Industry Transformation

- The IT industry transformation brings thinking on **network architecture** and **device architecture** in the network industry. The network architecture layer involves the SDN controller and the device architecture layer involves the device deployment mode.

IT Industry Transformation

- In recent years, IT technologies such as virtualization and cloud computing have been booming, and applications deployed on hardware have been gradually migrated to the cloud. Applications are deployed on private clouds, public clouds, or hybrid clouds as software.



- Thinking about the network industry: Can network applications be deployed in a software-based manner?
- In the context, Network Functions Virtualization (NFV) is introduced.



Origin of NFV

- In October 2012, 13 top carriers (including AT&T, Verizon, VDF, DT, T-Mobile, BT, and Telefonica) released the first version of NFV White Paper at the SDN and OpenFlow World Congress. In addition, the Industry Specification Group (ISG) was founded to promote the definition of network virtualization requirements and the formulation of the system architecture.
- In 2013, the ETSI NFV ISG conducted the first phase of research and completed the formulation of related standards. The ETSI NFV ISG defined NFV requirements and architecture and sorts out the standardization processes of different interfaces.





NFV Value

- NFV aims to address issues such as complex deployment and O&M and service innovation difficulties due to large numbers of telecom network hardware devices. NFV brings the following benefits to carriers while reconstructing telecom networks:
 - Shortened service rollout time
 - Reduced network construction cost
 - Improved network O&M efficiency
 - Open ecosystem



Key NFV Technologies: Virtualization

- Virtualization is the foundation of NFV, and cloudification is the key.
- On traditional telecom networks, each NE is implemented by dedicated hardware, resulting in high costs and difficult O&M. Virtualization features partition, isolation, encapsulation, and independence from hardware, which can meet NFV requirements. Carriers use virtualization to run software-based NEs on universal infrastructures.

Partition



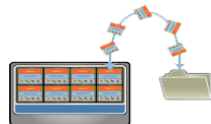
Multiple VMs can concurrently run on a single physical server.

Isolation



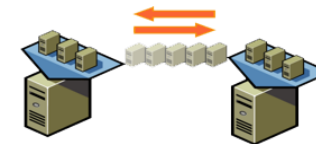
VMs that run on the same server are isolated from each other.

Encapsulation



All data of a VM is saved in files. A VM can be moved and replicated by moving and replicating the files.

Hardware independence



VMs can run on any servers without any modifications.



Key NFV Technology: Cloudification

- As defined by the National Institute of Standards and Technology (NIST), cloud computing is a model that allows users to obtain resources (for example, networks, servers, storage devices, applications, services) in a shared compute resource pool based on their needs anytime, anywhere. This model enables fast resource provisioning and release, and minimizes the resource management workload and interactions with service providers.
- Cloud computing has many advantages. Cloudification of network functions on carriers' networks mainly uses resource pooling and rapid elastic scaling.

Characteristics of Cloud Computing

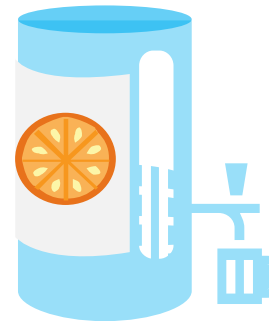
1 On-demand self-service



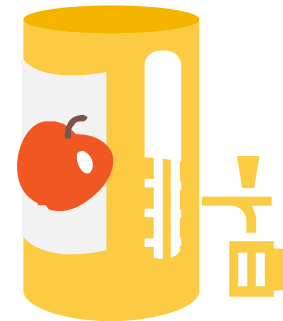
2 Broad network access



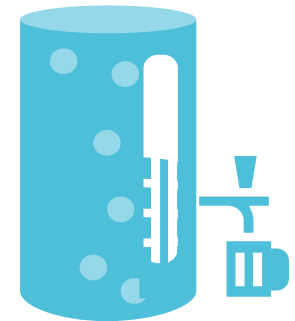
3 Resource pooling



4 Rapid elasticity



5 Measured service





Introduction to the NFV Architecture

- The NFV architecture includes the network functions virtualization infrastructure (NFVI), a virtualized network function (VNF), and management and orchestration (MANO). In addition, the NFV architecture needs to support the existing business support system (BSS) or operations support system (OSS).

OSS/BSS: is an existing operation/O&M support system.

VNF: uses cloud resources to construct software NEs.

NFVI: provides cloud-based resource pools.

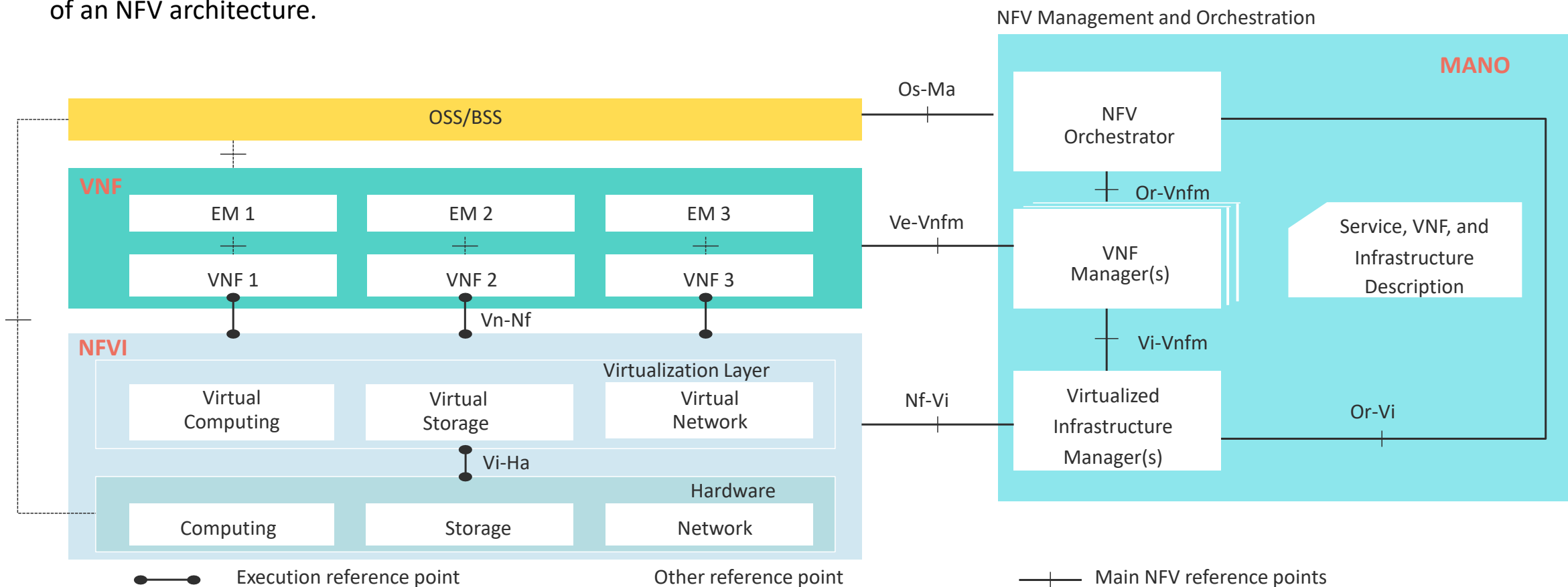
MANO:

Provides functions such as service orchestration, service management, and resource management.



Standard NFV Architecture

- ETSI defines the standard NFV architecture, which consists of the NFVI, VNF, and MANO. The NFVI includes the universal hardware layer and virtualization layer. The VNF is implemented using software, and the MANO implements management and orchestration of an NFV architecture.





Functional Modules of the NFV Architecture

- Main functional modules defined in the standard NFV architecture:

OSS or BSS

Management system for a service provider. It is not a functional component in the NFV architecture, but the MANO must provide an interface for interoperation with the OSS or BSS.

MANO

NFV management and orchestration. The MANO includes the VIM, VNFM, and NFVO, and provides unified management and orchestration for VNFs and the NFVI.

- **VIM:** NFVI management module that runs on an infrastructure site. The VIM provides functions such as resource discovery, virtual resource management and allocation, and fault handling.
- **VNFM:** It controls the VNF lifecycle (including instantiation, configuration, and shutdown).
- **NFVO:** It orchestrates and manages all the software resources and network services on an NFV network.

VNF

VNFs refer to VMs as well as service NEs and network function software deployed on the VMs.

NFVI

NFV infrastructure, including required hardware and software. The NFVI provides a running environment for VNFs.

- **Hardware layer:** includes hardware devices that provide compute, network, and storage resources.
- **Virtualization layer:** abstracts hardware resources to form virtual resources, such as virtual compute, storage, and network resources. The virtualization function is implemented by Hypervisor^[1].



NFV Architecture Interfaces

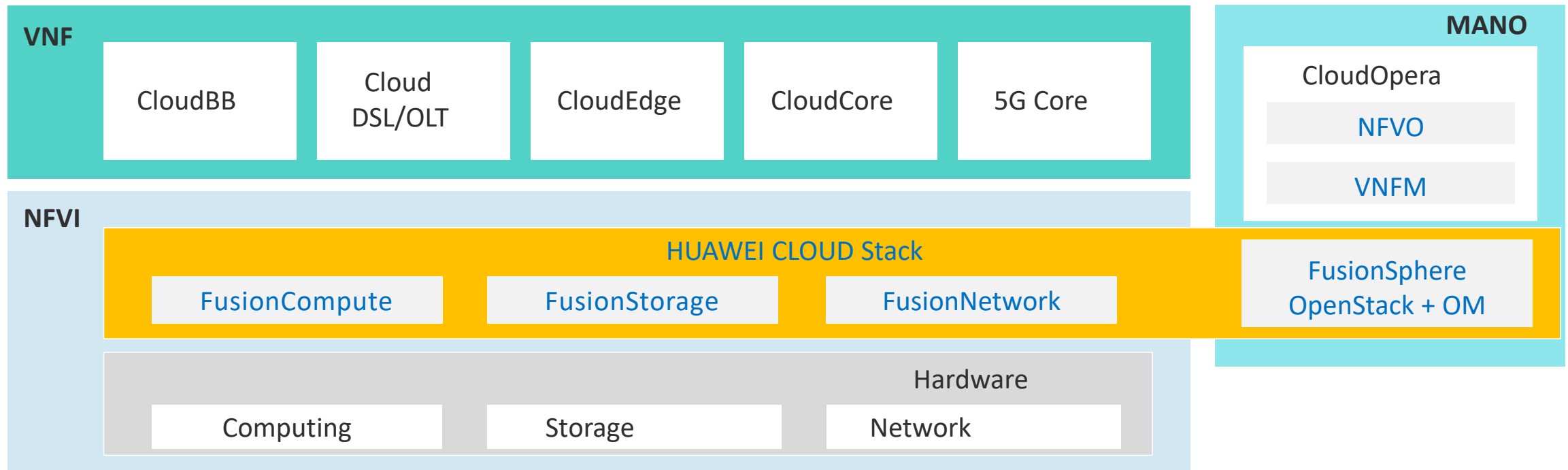
- Main interfaces of the standard NFV architecture:

Interface	Description
Vi-Ha	Is used between the virtualization layer and hardware layer. The virtualization layer meets basic hardware compatibility requirements.
Vn-Nf	Is used between a VM and the NFVI. It ensures that VMs can be deployed on the NFVI to meet performance, reliability, and scalability requirements. The NFVI meets VMs' OS compatibility requirements.
Nf-Vi	Is used between the virtualization layer management software and NFVI. It provides management of virtual computing, storage, and network systems of NFVI, virtual infrastructure configuration and connections, as well as system usage, performance monitoring, and fault management.
Ve-Vnfm	Is used between the VNFM and a VNF, implementing VNF lifecycle management, VNF configuration, VNF performance, and fault management.
OS-Ma	Manages lifecycles of network services and VNFs.
Vi-Vnfm	Is used for interaction between the service application management system or service orchestration system and virtualization layer management software.
Or-Vnfm	Sends configuration information to the VNFM, configures the VNFM, and connects the orchestrator and VNFM. It exchanges information with the NFVI resources allocated to VNFs and information between VNFs.
Or-Vi	Is used to send resource reservation and resource allocation requests required by the orchestrator and exchange virtual hardware resource configurations and status information.



Huawei's NFV Solution

- In the Huawei NFV architecture, functions of the virtualization layer and VIM are implemented by the HUAWEI CLOUD Stack NFVI platform. HUAWEI CLOUD Stack can virtualize compute, storage, and network resources and centrally manage, monitor, and optimize physical virtualization resources.
- Huawei provides cloud-based solutions for carriers' wireless networks, bearer networks, transport networks, access networks, and core networks.





FAQ

- Q1: What is the relationship between SDN and NFV in the industry?
- A: Both SDN and NFV involve network transformation and the NFV concept was proposed at the SDN and OpenFlow World Congress. However, they are independent of each other. SDN mainly affects the network architecture, and NFV mainly affects the NE deployment mode.
- Q2: What is the relationship between SDN and NFV in Huawei solutions?
- A: Huawei provides different solutions for SDN and NFV, but they are associated. Huawei NFVI solution is provided by HUAWEI CLOUD Stack.



Quiz

1. (Multiple) Which of the following statements about Huawei SDN solution are true? ()
 - A. The solution supports various SBI protocols, such as RESTful, NETCONF, and OVSDB.
 - B. OpenFlow can be used as the SBI protocol.
 - C. The solution integrates management, control, and analysis to build a simplified network.
 - D. The solution provides open and programmable network interfaces to support third-party application development and system interconnection.
2. Please briefly describe the benefits of NFV.



Summary

- With the transformation and development of the network industry, SDN and NFV are proposed.
- SDN is an innovation of network architecture. It uses a controller to make networks more open, flexible, and simple.
- NFV is an innovation in the deployment of telecom network devices. Based on virtualization and cloud computing, NFV helps reconstruct telecom networks.



More Information

- For more information about OpenFlow, visit <https://www.opennetworking.org/> .
- For more information about Huawei SDN solution, see the HCIP course.

The image features a blue-tinted background with silhouettes of several groups of business professionals in a modern office environment. The silhouettes are arranged in three main clusters: a group of four on the left, a group of four in the center, and a group of five on the right. They appear to be engaged in collaborative work, with some individuals holding documents or pointing towards something off-camera. The background shows architectural details like glass panels and structural beams, creating a sense of a high-rise office building. The overall mood is professional and corporate.

Thank You
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